A Closed Form for What the Model Emits: Template Anchoring in Unconditioned Zero-Shot Circle Packing

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Abstract

Language models placed in the proposal role of discovery loops such as FunSearch and AlphaEvolve are assumed to explore a solution space. We characterize the *unconditioned* proposal — a single zero-shot call with no parent program and no fitness feedback — on a constructive-geometry benchmark scored by an exact evaluator: maximize the sum of radii of N circles in a unit square. A weak-tier proposer does not search. It emits a grid template whose value is predictable in closed form from the problem parameters alone — a nearest-square order $k^* = \text{round}(\sqrt{N})$ with value function $V(k,m)$ — registered before sampling and modal at all seven N of the original square arm, and, in a held-out probe registered against a recall alternative, at four of five N absent from every cited scoreboard (all three discriminating cells; template structure at all five). Preregistered extensions bound the result: the anchor dissolves under parent conditioning, the regime discovery loops actually run in, so every claim is scoped to unconditioned calls. Handed the higher-valued construction, the model chooses it and fails to build it in 30 of 31 invalid attempts (the remaining one unevaluable): the template is not what the model prefers but what it can reliably execute. Delegating construction to an executed math-only program does not escape it: across three registered code-channel arms spanning two tiers, 0 of 115 valid program outputs exceed the family argmax and the anchor remains the modal weak-tier output — at these cells the ceiling is a property of the channel, not of the tier. The behaviour transfers to two further vendors at reduced concentration (61–67% of valid outputs against 56–76% in-family at the same cells) and is bracketed by a fourth family, one member escaping upward and the other anchoring nowhere: anchoring is not a single-vendor artifact, and its concentration is vendor-dependent.

1 Introduction

Ask a language model to place 21 circles in a unit square so as to maximize the sum of their radii, and it will not search. It reaches for the nearest square grid — five by five, radius $1/(2k)$ — then truncates it to 21 circles, leaving four cells empty and their area unclaimed. Those cells are where the value went. A better construction is one parameter away: drop to a four-by-four grid and add corner fillers of radius $(\sqrt{2}-1)/(2k)$, and the sum rises. The model does neither, and behaves the same way on the next sample.

The behavior can be written down. Writing $k^*(N) = \text{round}(\sqrt{N})$ for the nearest-square order, a value function $V(k,m)$ over grid order k and filler count m identifies the modal output of the proposal distribution — which template the model emits most often at each N, and hence its value, from the problem parameters alone, before sampling. The empirical mode equals the predicted value at all seven N tested, and per-sample agreement is bounded by the modal frequency itself, 56–86% by cell (§3.2). The rule also predicts where the behavior costs value: when $k^{*2} \leq N$ the model extends the grid with fillers (a branch arm §3.4 later restricts to $m \leq 1$); when $k^{*2} > N$ it truncates instead of dropping to k^*-1 and filling. These *trap zones* — $N \in [13,15]$, $[21,24]$, $[31,35]$, $[43,48]$, $[57,63]$ (clipped to $[57,60]$ by our sweep bound) — are a property of the branch behavior, not of the packing problem.

Circle packing is the showcase benchmark of the LLM-driven discovery literature, the lineage FunSearch (Romera-Paredes et al., 2024) opened. It is the task on which AlphaEvolve (Novikov et al., 2025) reported

2.63586276 for $n = 26$ and on which later systems report a narrow band — ShinkaEvolve (Lange et al., 2025) 2.635983283, HELIX 2.63598308 (Su et al., 2026), GigaEvo 2.636 (Khrulkov et al., 2025), AdaEvolve 2.636 (Cemri et al., 2026) — with ThetaEvolve (Wang et al., 2025) claiming new best-known bounds rather than matching it.

What we measure, and what we do not. Those systems query p(program | parent programs + fitness scores + evaluator feedback + archive context, at generation $t > 0$). We query p(coordinates | task description, no parent, no code channel, no feedback, generation 0). These differ on four axes at once — parent conditioning, output modality, feedback, generation index — and we do not claim to have measured the in-loop distribution. What we characterize is the *unconditioned* call. A source audit finds no loop stage in the cited systems that makes precisely this call: FunSearch reseeds emptied islands by cloning a best surviving program, ShinkaEvolve's stagnation restart re-seeds from its own archive ("initial", "best" or "archive_random"), and OpenEvolve islands copy the user-provided seed — every in-loop call is program-conditioned. The nearest deployed relatives are initializations conditioned on a seed that "can be trivial" (FunSearch) or "rudimentary", single-line functions returning constants (AlphaEvolve), plus AlphaEvolve's "No evolution" ablation, which re-prompts from the same rudimentary program throughout. The unconditioned call is the limiting case these approach as seed information goes to zero, and finding that this limit is a template lookup, its value computable in advance, is actionable for the people who build these loops — seed diversification, forced-k initialization, trap-N avoidance in benchmark selection. For an evolutionary-computation reader the consequence sits at initialization: if generation zero is one template wearing many coordinates, the diversity machinery downstream — MAP-Elites descriptors (Mouret & Clune, 2015), novelty pressure (Lehman & Stanley, 2011), seed perturbation — is doing repair rather than exploration. For an evaluation reader it is a floor: the weak-tier zero-shot mode on this benchmark is computable in advance, so a discovery claim on circle packing has a known value to beat before any search is credited. Anything about what the loop samples at $t > 0$ is a hypothesis here. The one published result bearing directly on the substitution — "Dictionaries, Not Darwin" (Li, 2026), reporting parent-conditioned evolution indistinguishable from fresh independent sampling at equal budget — is warrant from equation discovery, not geometry. §8 names the experiment that would close the gap.

This paper makes three framing choices worth stating plainly. Behavioral anchoring is established territory (Huang et al., 2026; Lou & Sun, 2024; Malberg et al., 2025) and we inherit its vocabulary; what differs is modality — the anchor is a *construction template*, not a number. The benchmark choice is strategic; we did not pick it for novelty. Preregistration, too, is an adopted standard here, not a contribution in itself (Thomas et al., 2026; Williams et al., 2026; Vaccaro, 2026; Zhang, 2026); our twist is what is locked — exact-output point predictions from a closed form, on a held-out container.

Contributions.

1. *A choice/execution dissociation: the model selects the better construction and cannot build it through either channel.* Handed the provably higher-valued in-family construction with its score, the weak tier chooses it, then fails to instantiate it in 30 of 31 invalid attempts, the remaining one unevaluable (arm CH, §3.5); delegated to an executed math-only program, the same ceiling holds across two tiers and a fresh-draw replication — 0 of 115 valid program outputs exceed the family argmax (arms CC/CC2/CCS, §3.7). The template is not what the model prefers but what it can reliably execute, in text or in code. This finding does not depend on whether any deployed loop runs unconditioned calls.

2. *A closed form that names the modal model output in advance.* $k^*(N)$ with $V(k, m)$ and $T(k, N)$ identify the value a weak-tier proposer emits most often at a given N, verified against a linear-programming oracle on 83 configurations to within 10^{-9} and tested out of sample on two containers, the rectangle rule restated ($q^* = \mathrm{round}(\sqrt{N/a})$, $p^* = \mathrm{round}(\sqrt{N \cdot a})$) but never refitted. Prior work publishes functional forms predicting *aggregate* metrics ahead of a run (Kaplan et al., 2020; Hoffmann et al., 2022; Snell et al., 2024); to our knowledge none predicts a specific multi-decimal *individual emitted output value* ahead of sampling. The claim is scoped to the weak tier (§5).

3. *A tier boundary condition, as three attractor families.* Ambition rises monotonically with nominal tier; validity does not — 83% → 100% → 13% at the primary tolerance on matched cells (78% over

all seven weak-tier cells), 64% (all seven weak-tier cells; the matched-cell weak rate at this tolerance was not computed) → 90% → 13% at 10^{-9} (Table 2). The most ambitious tier attempts recursive gaskets and mostly fails to produce a valid packing; it was addressed through a bare serving alias and appears as `opus_alias` with that caveat everywhere.

4. *Checkable faithfulness, and trace requests as interventions.* Method lines are checkable against emitted coordinates, and 54 of 56 scoreable claims (96.4%) describe the object actually built (Wilson 95% interval $[88\%, 99\%]$, §6.5). Requesting the line at all is an intervention rather than an observation: the concentration shift it produced (87% vs 70%, $p = 0.0325$ uncorrected) fails multiplicity correction, is carried by one of three cells and is wave-confounded (§6.4), so it is reported as met-as-registered exploratory material, not a demonstrated effect. Studies collecting process descriptors *by request* are nonetheless measuring a perturbed distribution.

5. *A family boundary, measured from both sides — and a near-deterministic strong form.* Anchoring is not a single-vendor artifact: it transfers to two further vendors' weak models at reduced concentration (61–67% of valid outputs against 56–76% in-family at the same five cells, §3.2 and §3.6), while one further family brackets it — gemma-4-26b escapes *upward* to the in-family rival construction (27/43 discriminating-cell validities) and gemma-4-31b, above the same validity floor, anchors on nothing (§3.6). Where anchoring holds, its strong form is near-deterministic: pooling every weak-tier discriminating-cell validity across the original-family and transfer-vendor arms, the provably better in-family rival is emitted 3 times in 186 (arm F 2/57, trace 1/53, rectangle 0/11, arm M 0/10, arm V 0/16, arm CN 0/39 — a descriptive pool over separately registered weak-tier arms, each reported at its own bar in §3–§4 and §6; the Sonnet tier reaches the rival's structure 5 times in 30 and is excluded by tier, §5; gemma-4-26b, the family that does *not* anchor at its discriminating cells, is the counterexample that proves the statistic is a family trait, §3.6).

Non-claim guard. Everything here is a behavioral regularity over emitted outputs. We make no claim about mechanism inside the weights, and the paper uses that vocabulary throughout: the model *emits*, the distribution *concentrates*, the formula *identifies the modal output* (storage-and-retrieval vocabulary corrected). §8 states the leading mechanism-free alternative, arithmetic tractability, and reports its direct probe (arm CH, §3.5): wrong about selection, right about instantiation.

1.1 Trap zones

Substituting $k^*(N)$ into the branch condition: TRAP $N \in [k^2 - k + 1, k^2 - 1]$, CONVERGE $N \in [k^2, k^2 + k]$. Over $N = 10 \dots 60$: $[13, 15]$, $[21, 24]$, $[31, 35]$, $[43, 48]$ and $[57, 63]$ clipped to $[57, 60]$ by the sweep bound.

Inside a zone the rule generally loses value against the best construction available *within the recipe family*. Worst-in-zone penalties fall as N grows: 8.51% at $N = 13$, 7.03% at 21, 6.01% at 31, 5.25% at 43, 4.66% at 57. The penalty reaches zero at $N = 35$, 48, 14 and 15, but **not** at $N = 24$, where truncation gives $T(5, 24) = 2.4000000$ against $V(4, 8) = 2.4142136$, a residual 0.59%. Branch label and penalty are therefore not coextensive: 4 of the 22 trap-branch N in range cost nothing, so "trap" names the branch, not a guaranteed loss. Where the penalty is zero, truncation is separable from convergence only by structure; §3.3 uses $N = 35$ as that control.

Every closed-form value is recomputed over the constructed coordinates by an independent linear program that knows nothing about the recipe; the script aborts on disagreement. 83 configurations, both branches, every k in $2 \dots 7$, drift below 10^{-9}.

Our bound table (`n_sweep_forecast.json`, transcribed from Friedman (n.d.)) covers $N = 10 \dots 30$ only, and its $N = 26$ entry, 2.63598, *is* ShinkaEvolve's figure truncated, so the LLM-driven systems are the record on this problem. What survives is that the recipe family is never competitive with that record: deficit 0.02–0.26 across $N = 10 \dots 30$, and 0.0946 at $N = 26$ ($2.63598 - 2.5414214 = 0.0945586$). Above $N = 30$ there is no bound table, so both the deficit claim and the LP gate's "never exceeds a published bound" abort are unchecked there, covering three of five trap zones and four of our seven square cells.

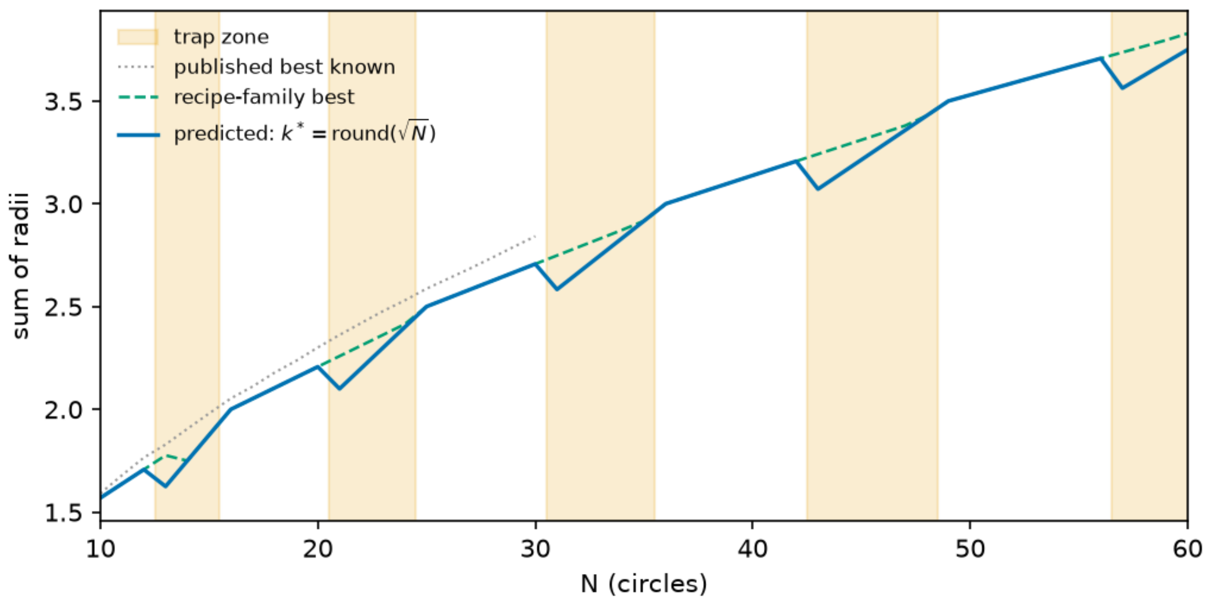

Figure 1: *Predicted versus optimal value,* $N = 10 \dots 60$. (i) the rule's prediction; (ii) the best value in the recipe family; (iii, dotted) published best-known values (Friedman, n.d.), terminating at $N = 30$ because the bound table stops there. Shaded bands are the trap zones, the last being $[57, 63]$ clipped by `sweep(10,60)`. Worst-in-zone penalties 8.51%, 7.03%, 6.01%, 5.25%, 4.66%; these are family-internal arithmetic and exact at every N, while the published-bound comparison (curve iii) is checkable only up to $N = 30$, where the bound table stops. The gap closes to zero at $N = 35$ and $N = 48$ but **not** at $N = 24$, where 0.59% remains — curve (ii) sits strictly above (i) across $[21, 24]$.

2 Task and recipe family

2.1 The benchmark

Maximize the sum of radii of N non-overlapping circles in the unit square. Feasibility is decidable exactly from emitted coordinates, so proposals score with no human in the loop.

All invocations in the core arms are zero-shot and code-free. The prompt (Appendix A) fixes the count, states containment and non-overlap, forbids writing or executing code, and demands only a raw Python list of `[x, y, r]` triples. Our reason for the code-free design is that a code channel routes the task to an executed optimizer, so the distribution would reflect the optimizer. That reason is now measured rather than asserted: three preregistered code-channel arms (§3.7 — weak tier, a fresh-draw replication, and a Sonnet-tier probe) find that delegating construction to a math-only executed program neither escapes the family ceiling (0 of 115 valid outputs above the argmax, across both tiers) nor dislodges the anchor as the modal weak-tier output, so the code-free scope costs less loop relevance than we had assumed — with the library-assisted channel (`numpy`/`scipy`), the one deployed loops permit, still unmeasured.

2.2 The recipe family

Three terms, used consistently: the **recipe family** is the parametric set of grid-plus-filler constructions; a **template** is one member; the **anchor** is the template the distribution concentrates on at a given N.

A $k \times k$ grid places one circle per cell center with $r_{\text{grid}} = 1/(2k)$; m fillers sit on interior grid vertices, tangent to their four surrounding grid circles, with $r_{\text{filler}} = (\sqrt{2} - 1)/(2k)$, $0 \le m \le (k-1)^2$. Hence $V(k, m) = k/2 + m(\sqrt{2} - 1)/(2k)$. When $N < k^2$ the observed behavior is not to drop to a smaller grid and fill it but to *truncate* — lay out the $k \times k$ lattice, occupy N cells, leave the radius unchanged — giving $T(k,N) = N/(2k)$. These reproduce every previously reported anchor with no fitting: $V(5,1) = 2.5414214$, $V(5,2) = 2.5828427$, $V(5,0) = 2.500$, $T(5,23) = 2.300$, $V(4,7) = 2.3624369$.

The finding that 94 of 95 valid proposals in prior arms were grid-plus-filler is retained only as motivation, since those arms are not reported here; the self-contained version is §5's observation that across the 155 weak-tier rows collected before arm M only two exceeded the family best, both by $\sim 10^{-7}$ (arm M's 57 rows are scored separately in §3.4 and were not swept for this claim).

2.3 The order rule is a definition; the branch rule is the hypothesis

We write $k^*(N) = \lfloor \sqrt{N} + \frac{1}{2} \rfloor = \mathrm{round}(\sqrt{N})$ for the nearest-square order. This is a *definition*. Given that the model emits some $k \times k$ grid from this family, "the nearest square lattice" yields $\mathrm{round}(\sqrt{N})$ by construction; and $\arg\min_k |k^2 - N|$ switches at $N = k^2 + k + \frac{1}{2}$ where $\mathrm{round}(\sqrt{N})$ switches at $k^2 + k + \frac{1}{4}$, so on the integers the two coincide exactly — two formalizations of "nearest" agreeing is a symptom that no substantive hypothesis is being selected. Scoring four candidate orders against the three fitting anchors gave `nearest` 3/3, `floor` 2/3, `argmax` 2/3, `ceil` 1/3; under a null where each candidate matches each anchor with probability $\frac{1}{2}$, some candidate among four goes 3/3 about 40% of the time, so this is a weakly identified discrete selection, not a zero-parameter law.

The falsifiable content is the **branch rule**: $k^{*2} \le N \to$ extend with $m = N - k^{*2}$ fillers (converge); $k^{*2} > N \to$ truncate the k^*-grid (trap). The extend arm survived testing only at $m \le 1$ — a preregistered extension disconfirmed it at $m \ge 4$, where the model prefers the (k^*+1) rectangle grid (§3.4). Nothing about "nearest" forces truncation — a proposer with any local search would drop to k^*-1 and fill, which is why the trap zones exist and why a higher-valued rival is reachable inside them. Truncate-versus-drop-and-fill is a genuine dichotomy, and it is what §3 and §4 test. The governing quantity is the signed distance $N - k^{*2}$. Primality is not the variable — 13, 23, 31, 43, 47 and 59 trap while 11, 17, 19, 29, 37, 41 and 53 converge — refuting the prior work's conjecture about primes near 30 before any model is queried.

2.4 Notation and conventions

Terms used throughout, gathered here so no verdict table depends on a definition made elsewhere:

Term	Meaning
$k^*(N)$	nearest-square grid order, $\mathrm{round}(\sqrt{N})$ (a definition, §2.3)
$V(k,m)$, $T(k,N)$	extend-with-m-fillers value $k/2 + m(\sqrt{2}-1)/(2k)$; truncated-grid value $N/(2k)$
anchor	the template the proposal distribution concentrates on at a given N
attractor	the same object, used when the emphasis is on the family a tier converges to rather than on one cell (§5)
on-prediction	emitted sum within 2×10^{-3} of the registered predicted value
rival (argmax)	the highest-valued member of the recipe family at that N when it differs from the prediction — inside a trap zone, $V(k^*-1, m)$
discriminating cell	N where prediction $\neq$ family argmax; only these can distinguish anchoring from family search
validity	non-overlap and containment at 10^{-6} (primary) and 10^{-9} (logged), both always reported
trap / converge	branch by sign of $N - k^{*2}$: truncate when $k^{*2} > N$, extend when $k^{*2} \le N$
tier	nominal capability class of the serving alias addressed (weak/mid/top); `opus_alias` is a bare alias with no attestable weights binding
BELOW-FLOOR	model failing its arm's registered validity floor; no anchoring claim in either direction
UNSCOREABLE	cell with <3 valid samples under the GM-chain registration; claims nothing
mixed-serving accounting	pooled rate including cells collected through a registered serving-path amendment; the first-party-only rate is always given beside it

3 Preregistered forecast, out of sample

Sixteen registered arms appear in this paper. Each was registered before its sampling, each is reported regardless of outcome, and the falsifiers that fired are flagged in place:

Arm	§	Proposer(s)	Design	Registered test	Outcome
F (square, bare)	3.2–3.3	weak tier	7 N × 5–20	P1–P5 exact-value modes	modal at 7/7 N
S (mid tier)	5	Sonnet tier	3 N × 10	P-S1–P-S4 tier contrasts	1/30 on-prediction, 100% valid
O (top alias)	5	`opus_alias`	3 N × 10	P-O1–P-O4	13% valid; 3 predictions not evaluable (deviation, Table 4)
T (trace)	6	weak tier	3 N × 20 × 2 arms	P-T1–P-T4 + falsifier	falsifier **triggered** (tie reading); P-T3 fragile
G (rectangle)	4	16 proposers	2 cells	out-of-sample transfer	partial, 5/11; formula negative result kept
M (branch stress)	3.4	weak tier	4 N × 15	P-M1–P-M4 + F-M1/F-M2	F-M1 **triggered 3/3**: extend branch dies at $m \geq 4$; fifth k confirms
MU (mutation)	3.5	weak tier	3 parents × 3 N × 15	P-MU1–P-MU4 + F-MU1	F-MU1 **triggered**: anchor dissolves under conditioning
CH (choice)	3.5	weak tier	3 N × 15	P-CH1 vs P-CH2	choice follows score table; execution fails off-template 30/31
GM (gemini-flash-lite)	8	2nd vendor	7 N × 20	GM-chain bars	UNDERPOWERED (20 valid / 140)
GM2 (gemma, 4k budget)	8	2nd vendor	7 N × 20	GM-chain bars	0/140 parseable — budget confound, diagnosed by GM3
GM3 (gemma, 16k budget)	3.6	gemma-4-26b	7 N × 20	P-GM3.1–.3 + falsifier	pooled bar met; misses are the rival construction
V (cross-vendor zero-shot)	3.6	13 free-tier aliases	5 N × 5	floor + V1/V2	2 vendors TRANSFER, gemma-4-31b DOES-NOT-TRANSFER, V2 HOLDS
CC (code channel)	3.7	weak tier	3 N × 15	P-CC1 vs P-CC2	P-CC1 holds: anchor stays modal; 0/37 above the family argmax
CC2 (fresh-draw replication)	3.7	weak tier	3 N × 15	R1 + R2 + F-CC2.1	**REPLICATED**: 0/41 above argmax, anchor modal 3/3
CCS (tier probe)	3.7	Sonnet tier	3 N × 15	P-CCS1 vs P-CCS2	P-CCS2 holds: 0/37 above argmax — ceiling is channel, not tier
CN (held-out N)	3.8	weak tier	5 N × 15	P-CN1 vs P-CN2 + S-CN1–3	P-CN1 holds: mode at 4/5 held-out cells, 3/3 discriminating; structure 5/5

3.1 Registration protocol

For each cell the prompt is pinned and its SHA-256 digest written to disk first — the $N = 13$ square prompt hashes to `32db485b...`. Predictions P1–P5 are registered in the header of `arm_f_repro.py`, with standalone files for the parallel arms (`arm_s_preregistration.txt`, `arm_o_preregistration.txt`, `arm_t_preregistration.txt`). Raw outputs are stored verbatim, failures included; scoring is deterministic and local.

Four properties here are disclosed, not repaired. Temperature, top-p and top-k are not exposed by the runtime. The alias-to-weights binding is a promise, not a hash. The subagent inherits instruction files outside the task prompt, so the digests lock a *fragment* of the conditioning context — they are prompt-fragment hashes, and a replicator cannot reconstruct the inherited files from what we release. Finally, hash coverage is incomplete (§9).

Two scoring conventions were fixed in advance. Validity is reported at 10^{-9} and 10^{-6}, both logged, 10^{-6} primary: proposers print six to eight decimals, so an eight-decimal tangency misses contact by $\sim 5 \times 10^{-9}$, while 10^{-6} sits far below the $\sim 10^{-2}$ gap between rival constructions. Both tolerances appear in every validity figure below, as registered. Value matching uses a 2×10^{-3} window; §5 records where that window is too loose for the distinction it was asked to carry.

> **Reproducibility posture.** Prompt hashes were registered before sampling in the preregistration files for arms F, S, O and T (`arm_f_repro.py` header, `arm_s_preregistration.txt`, `arm_o_preregistration.txt`, `arm_t_preregistration.txt`); what is hashed is the task prompt only, not the harness, so the dispatch wrapper and the inherited instruction files sit outside the digest; what is not attestable at all is the `opus_alias` alias-to-weights binding, a vendor promise rather than a hash; hash-coverage gaps, the unhashed pilot prompt and the ledger provenance corrections are itemized in §9 and Appendix A.

3.2 The formula sits at the mode ceiling

An exact point prediction invites a fair objection: a per-sample hit rate well short of 100% is not "predicting the exact output". The number should be read against the sampling entropy of the distribution, computed in `p11_mode_baseline.json` (bare arm, 2×10^{-3} buckets). At **all seven tested N the predicted value is the empirical modal output** — four discriminating cells plus three where the prediction and the family argmax coincide, so the anchoring-specific evidence sits in the four — and the on-prediction rate equals the modal frequency to within one sample:

N	predicted	empirical mode	modal freq	on-prediction / valid	k^*-structure / valid†
13	1.6250000	1.6250	10/18	10/18 (56%)	17/18
17	2.0517767	2.0518	3/4	3/4 (75%)	3/4
21	2.1000000	2.1000	12/15	12/15 (80%)	13/15
31	2.5833333	2.5833	12/17	13/17 (76%)	17/17
35	2.9166667	2.9167	3/4	3/4 (75%)	3/4
37	3.0345178	3.0345	3/4	3/4 (75%)	4/4
43	3.0714286	3.0714	6/7	6/7 (86%)	7/7

† Post-hoc structural check, not preregistered, run after the square arm closed (`diagnostics_kmatch.py`): the grid order backed out of each sample's dominant radius ($k = \mathrm{round}(1/2r)$) equals $k^*(N)$. All 50 of 50 on-prediction samples are k^*-structured — the value match is not an accounting coincidence — and 64 of 69 valid samples overall (93%) sit on the k^* grid, so most value misses are k^*-grid variants (perturbed radii or filler counts), not different constructions.

No predictor of a single value can exceed the modal frequency, and this one attains it at 7 of 7 cells (at $N = 31$ the on-prediction count exceeds the 4-dp modal bucket by one sample — 13/17 against 12/17 — a bucketing convention, not a violation of the ceiling): the residual is dispersion, not misprediction. Three cells ($N = 17$, 35, 37), all non-discriminating, carry four valid samples each, where a 3/4 mode is thin evidence on its own; the well-populated cells of the held-out probe (§3.8) are the stronger test of the same rule. As a floor, a naive "the model emits a round number" baseline hits 2 of the same 69 valid samples (3%). Two scope conditions: this is the bare weak-tier arm — pooled over the four discriminating square cells (full ledger), the two higher tiers and the rectangle, the on-prediction rate is 46% ($47/102 = 41/57 + 1/30 + 0/4 + 5/11$; the three non-discriminating square cells, 9/12, are excluded because prediction equals the family argmax there), because §5's higher tiers are almost never on-prediction — and "modal" is a property of the observed sampling regime, not of a pinned decoding configuration (§8).

3.3 Square container

The square arm sampled Haiku-tier proposers at $N = 13$, 17, 21, 31, 35, 37 and 43. The rule was fitted on $N = 23/26/27$ only, so every cell is out of sample. Four cells are *discriminating*: rule and family optimum disagree, so a proposer that searched the family would return a higher number. Across those cells ($N = 13$, 21, 31, 43), **18 of 23 valid invocations landed on the predicted construction** and the rival-argmax value was reached **2 times in 23** — both at $N = 21$, both the 4×4 grid plus five fillers at 2.2588835.

Those figures cover the **original 45-invocation bare arm only** (ids ≤ 5 at $N = 13/31$, ≤ 10 at $N = 21$, all rows at $N = 17/35/37/43$), the corpus that existed when §3 was written. §6 later added 40 bare rows at $N = 13/21/31$ under the same arm label; over the full shipped ledger the same four cells give **41/57 on-prediction and 2/57 rival**. Both are reported. §3 and §6 are computed on nested subsets of one arm, not on independent samples, and Table 1 carries the split.

That the anchor first breaks at $N = 21$ is suggestive — it is the bottom of a zone carrying a 7.03% penalty, so the anchor looks weak where obeying it costs — but that is a hypothesis, not a finding. **P4** registered $N = 37$ as converging on 3.0345178, a prime predicted clean, contradicting the prior work's guess; it confirmed. **P5** registered $N = 35$ at 2.9166667, the top-of-zone control; three of four valid samples landed there, separable from convergence only because the structure classifier reads radii rather than totals; the fourth used a 7×7 lattice truncated to 35 (sum 2.5), outside the rule. At $n = 4$ this is "consistent with" rather than confirmed, and deserves running to $n = 20$.

Bookkeeping. Five invocations were rejected by the runtime's 20-subagent concurrency cap *before reaching a model*. We excluded them under a rule that was not preregistered, so the arm is reported both ways — $35/45 = 77.8\%$, or $35/50 = 70.0\%$ had the rejections been scored as model failures — and the five records appear in the ledger in no form (§9). The arm also contains three parse failures, and one $N = 37$ proposer derived $r = (\sqrt{2} - 1)/12$ correctly in prose, then transcribed 0.03571429.

3.4 Arm M: the filler branch fails at $m > 1$; the fifth k confirms

Two gaps remained after the square arm: (i) $V(k, m)$'s filler branch was empirically supported only at $m \in \{0, 1\}$ — the converge cells $N = 17$ and 37 both have $m = 1$ — and (ii) the $k = 8$ trap zone appeared in Figure 1 but was never sampled. Arm M closes both gaps and was registered before sampling (`arm_m_preregistration.txt`, commit `e528c6b`; prompt hashes in `arm_m_prompts.json`): $n = 15$ bare weak-tier invocations at each of $N = 20$ ($k^* = 4$, $m = 4$, predicted $V(4, 4) = 2.2071068$), $N = 30$ ($k^* = 5$, $m = 5$, $V(5, 5) = 2.7071068$), $N = 41$ ($k^* = 6$, $m = 5$, $V(6, 5) = 3.1725890$) and $N = 57$ ($k^* = 8$, trap, $T(8, 57) = 3.5625000$, rival $V(7, 8) = 3.7366935$), with an explicit tie-stated falsifier for each half.

The falsifier for the filler branch triggered at 3 of 3 cells. The registered predictions P-M1–P-M3 are disconfirmed: not one of the three converge cells has $V(k^*, m)$ as its modal output. The model does not extend a k^*-grid with fillers; it moves *up* to the (k^*+1) grid and truncates or exactly fills it — modal outputs $2.0 = T(5, 20)$ (12 of 15 valid; $N = 20$ is the 5×4 rectangle exactly), $2.5 = T(6, 30)$ (11 of 14; 6×5 exactly) and $2.9285714 = T(7, 41)$ (6 of 7; 7×6 minus one). Exactly one sample in 36 valid converge-cell rows emitted the registered construction — a textbook $V(4, 4)$, all four fillers at $(\sqrt{2} - 1)/8$. Per the registered falsifier wording, $V(k, m)$ **is restricted to the $m \leq 1$ support previously observed**, and the branch rule's extend arm is disconfirmed at $m \geq 4$; the intermediate "places some fillers" outcome named in the registration did not occur (filler count 0 in 35 of 36 valid converge rows). The emergent regularity is arithmetic, not geometric: where N factors as $a \times b$ with a, b near $\sqrt{N}$ the model emits that exact rectangle of uniform circles ($20 = 5 \times 4$, $30 = 6 \times 5$), which is further evidence for §8's arithmetic-tractability alternative — rectangle factorization is mentally computable where filler tangency is not.

P-M4 confirmed. At $N = 57$ the modal valid output is $T(8, 57) = 3.5625000$ (6 of 10 valid at 10^{-6}), the $k = 8$ grid truncated to 57; 0 of 10 valid samples reached the rival $V(7, 8)$, against a registered rival prediction of 0. The branch rule's truncate arm now has five confirmed k (4, 5, 6, 7, 8) and Figure 1's $k = 8$ zone is sampled. Validity by cell: 15/15, 14/15 (one parse failure — fraction literals), 7/12 and 10/15 at 10^{-6}. Three $N = 41$ invocations died in the runtime before reaching a model and are excluded under the rejection rule registered for this arm in `arm_m_preregistration.txt` — registered precisely because the analogous arm F exclusion was not (§9) — and counted here: the cell is $n = 12$ sampled of 15 launched. Raw rows verbatim in `arm_m_collect.jsonl`; scoring in `arm_m_analysis.py`, conventions identical to arm F.

Table M — Arm M, forecast versus outcome (all four registered cells).

N	k^*	Branch	Registered prediction	Modal valid output	Freq	Valid / sampled	Verdict
20	4	extend, $m = 4$	$V(4,4) = 2.2071068$	$T(5,20) = 2.0000000$ (5×4 exact)	12/15	15/15	P-M1 **disconfirmed**
30	5	extend, $m = 5$	$V(5,5) = 2.7071068$	$T(6,30) = 2.5000000$ (6×5 exact)	11/14	14/15	P-M2 **disconfirmed**
41	6	extend, $m = 5$	$V(6,5) = 3.1725890$	$T(7,41) = 2.9285714$ ($7 \times 6 - 1$)	6/7	7/12	P-M3 **disconfirmed**
57	8	truncate	$T(8,57) = 3.5625000$, rival count 0	$T(8,57) = 3.5625000$	6/10	10/15	P-M4 confirmed; rival 0/10

Exactly one $V(k^*, m)$ construction appears in 36 valid converge-cell rows; filler count is 0 in 35 of 36. Regenerates from `arm_m_collect.jsonl` via `arm_m_analysis.py`.

The net effect on the paper's claim is a sharpening. Arm M does not rescue $V(k,m)$; the closed form now reads "$k^* = \text{round}(\sqrt{N})$ names the grid family; at $k^{*2} > N$ the model truncates that grid (five k confirmed); at $k^{*2} \leq N$ it extends only to $m \leq 1$ — beyond that it prefers the $(k^*{+}1)$ rectangle, and $V(k,m)$ does not describe it."

3.5 Arms MU and CH: no transfer to one-parent calls; the argmax in hand does not dislodge the template

Two questions were still open after arm M. Does the unconditioned anchoring result say anything about calls that carry a parent program — the regime evolutionary loops actually run in — and does the §8 arithmetic-tractability alternative survive a direct probe? Arms MU and CH were registered together before sampling (`arm_mu_preregistration.txt`, commit `2b7d202`; prompt hashes in `arm_mu_prompts.json`), with numeric decision thresholds, a power analysis (0.83 at the registered effect size), tie-against conventions, and a falsifier (F-MU1) whose consequence — the title and abstract of this paper are scoped to *unconditioned* calls only — was written down before the first invocation.

Arm MU issues one-parent mutation calls: the bare prompt plus an existing packing and the instruction to modify it so the sum of radii increases. Three parents per cell ($N = 13$, 21, 31; $n = 15$ per condition-cell, 135 invocations): *A_anchor*, the model's own unconditioned anchor $T(k^*, N)$; *B_rival*, the provably better in-family rival $V(k^*{-}1, m)$; and *C_offfamily*, a diagonal-chain packing scoring ≈ 0.59–0.62, far below either.

F-MU1 triggered; the anchoring result dissolves under conditioning. Under B_rival the pooled anchor-rate is 0 of 26 valid (0%, CI $[0.00, 0.13]$) against a registered DISSOLVES threshold of $\leq 20\%$; keep-or-improve is 26 of 26 against a threshold of $\geq 50\%$. Every valid B_rival sample keeps or refines the rival construction, and every one inherits the rival's grid order (k-inheritance 26/26, disconfirming P-MU3's $< 50\%$ prediction). The template does not reassert itself when a better in-family parent is present. A_anchor makes the same point from the other side: even with the model's *own* anchor as parent and an explicit increase-the-sum instruction, only 7 of 28 valid samples (25%, CI $[0.13, 0.43]$) stay on the anchor — P-MU1's $\geq 50\%$ prediction is not met. Per the registered falsifier wording: the unconditioned-call anchoring result does not transfer to parent-conditioned calls, the loop-relevance framing is disconfirmed for the one-parent regime, and this paper's claims are scoped to unconditioned calls throughout.

The one condition where the template does reassert itself is the one the registration predicted it would: given the *off-family* diagonal parent, 20 of 34 valid samples (58.8%, CI $[0.42, 0.74]$) abandon the parent entirely and emit a k^*-grid construction — P-MU4 holds per-cell at $N = 21$ (7/13) and $N = 31$ (12/14, every valid sample on the $k = 6$ grid), though not at $N = 13$ (1/7). The asymmetry is the finding: a good in-family parent is followed, a bad off-family parent is discarded for the template. The attractor is real but weak — it captures the trajectory only when the parent is worse than what the template supplies.

Arm CH hands the model the entire recipe family: the bare prompt plus every admissible $T(k,N)$ and $V(k,m)$ value enumerated, including the family argmax, with the instruction to use whichever scores highest or anything better ($n = 15$ per cell, 45 invocations). The §8 tractability alternative says the model truncates at k^* because that is the only construction it can compute; enumerating the family removes the computation burden, so under that reading the modal output should become the stated argmax (P-CH2). Under the template reading it stays $T(k^*, N)$ despite the argmax being printed in the prompt (P-CH1).

Registered verdict: P-CH1 holds (2 of 3 cells) — but the registered metric carries a survivorship artifact we disclose rather than exploit. Under the registered scorer, which takes the modal output *among valid samples*, the template reading wins: at $N = 13$ the modal valid output is $T(4, 13) = 1.6250000$ against a stated argmax of 1.7761424, at $N = 31$ it is $T(6, 31) \approx 2.5833$ (4 of 7 valid) against 2.7485281 (3 of 7) — a one-sample margin — and only $N = 21$ goes to the argmax (2.2588835, 3 of 5 valid). P-CH2 holds at only 1 of 3 cells. Read no further and the tractability alternative fails the probe. But validity is low in this arm — 2, 5 and 7 valid of 15 by cell — and the failure modes reverse the picture at the *attempt* level: **all 31 invalid rows attempt the stated argmax** (every one carries the family's filler radius). Of those, 30 misplace the fillers — typically at the container corners, where radius $(\sqrt{2}-1)/(2k)$ overlaps the adjacent grid circle; the correct construction places them at the grid interstices — and 1 emits radical literals the registered scorer cannot evaluate. Counting attempts rather than valid outputs, the model *chooses* the argmax in a large majority of CH invocations and simply fails to execute it; the valid-conditioned modal is dominated by the template because the template is the only construction the model executes reliably. The registered metric was fixed before sampling, so we report its verdict as registered — and flag in the same breath that it conditions on exactly the variable (execution success) that separates the two readings. The honest summary is a decomposition rather than a clean winner: handed the argmax, the model's *choice* follows the score table, but its *execution* fails off-template in 30 of the 31 invalid attempts — the §8 tractability alternative is wrong about selection and right about instantiation. This is the same asymmetry arm MU found from the other direction, and it narrows the paper's central claim to its defensible core: the template is not what the model prefers; it is what the model can build.

Table MU — Arm MU, registered thresholds versus outcome, by parent condition (pooled over $N = 13$, 21, 31).

Parent	Valid	Anchor-rate (CI)	Registered threshold	Keep-or-improve	k-inher.	Reading
B_rival (in-family argmax)	26	0/26 (0%, $[0.00, 0.13]$)	DISSOLVES if $\leq 20\%$	26/26 (bar $\geq 50\%$)	26/26	**F-MU1 triggered** — anchoring does not transfer to one-parent calls
A_anchor (model's own anchor)	28	7/28 (25%, $[0.13, 0.43]$)	P-MU1 predicted $\geq 50\%$	—	—	not met — even the anchor-as-parent does not hold the anchor
C_offfamily (diagonal, $\approx$0.6)	34	20/34 abandon parent for k^*-grid (58.8%, $[0.42, 0.74]$)	P-MU4: template reasserts	—	—	holds at $N = 21$, 31; not at $N = 13$ (1/7)

Table CH — Arm CH, valid-conditioned verdict versus attempt-level decomposition.

N	Stated argmax	Modal valid output	Valid	Invalid rows attempting argmax	Registered verdict
13	1.7761424	$T(4, 13) = 1.6250000$	2/15	all	P-CH1 holds
21	2.2588835	argmax, 3/5	5/15	all	P-CH2 (argmax)
31	2.7485281	$T(6, 31) \approx 2.5833$ (4/7)	7/15	all	P-CH1 holds (one-sample margin)
pooled	—	template 2 of 3 cells	14/45	**31/31** (30 misplaced fillers, 1 unevaluable radicals)	choice follows score table; execution fails off-template

Raw rows verbatim in `arm_mu_collect.jsonl` (180 of 180 launched invocations collected across arms MU and CH, no runtime deaths); scoring in `arm_mu_analysis.py`, committed before sampling completed; frozen output in `arm_mu_results.txt`. This arm brings the study corpus to 468 invocations.

3.6 Arms GM3 and V: transfer at reduced concentration, with a boundary that runs both ways

Every arm above addresses one vendor's models. Two registered arms take the same byte-identical prompts across vendors — arm GM3 under a laboratory-grade instrument against one second-vendor family, arm V under free-tier serving against thirteen aliases spanning five vendors. One table carries the chain's verdicts; the two instruments are then reported in full.

Table CV — Cross-vendor verdicts, both instruments.

Vendor	Model	Instrument	Registered test	Verdict
Google	gemma-4-26b	first-party API, 7 $N \times 20$	pooled $\geq 30\%$; cell-level ≥ 5; falsifier ≥ 4 fails	pooled bar met (57.5%); discriminating cells 11/43 = 26%, **below** the bar; cell-level short (2/5, falsifier not triggered); all three misses are the *rival construction* (27/43)
Cohere	north-mini-code	free-tier alias, 5 $N \times 5$	V1 $\geq$ 50% of valid on-prediction; V2 rival $\leq$ 10%	**TRANSFERS**, point-estimate (8/12 = 67%; CI includes bar); V2 HOLDS (0/4)
OpenAI (open-weight)	gpt-oss-20b	free-tier alias, 5 $N \times 5$	same	**TRANSFERS**, point-estimate (11/18 = 61%; CI includes bar); V2 HOLDS (0/9)
Google	gemma-4-31b	free-tier alias, 5 $N \times 5$	same	**DOES-NOT-TRANSFER** (0/15); V2 HOLDS (0/3)
10 further aliases	(5 vendors)	free-tier	floor: $\geq$ 10 of 25 invocations valid	BELOW-FLOOR; failure taxonomies logged, no claim in either direction

Arm GM3. Arm GM3 asks whether the anchoring law is a fact about the original lineage or about weak-tier models as a class, by re-running the unconditioned-call protocol against a second vendor's open-weight family (gemma-4-26b-a4b-it) through its first-party API: seven N cells, twenty samples each, registered predictions and falsifier fixed before sampling (prereg chain in §9; the instrument's history — a first attempt whose 4,096-token budget was consumed entirely by the model's deliberation channel, resolved by a registered re-run at 16,384 — is in §8, and is itself a finding about output-budget confounds in cross-vendor comparisons). Per cell, recomputable from `arm_gm_gm3_checkpoint.jsonl` by `arm_gm3_analysis.py` with no arguments:

N	predicted (4 dp)	Gemma modal sum	mode freq	valid n	on-prediction	MODE-MATCH
13	1.6250	1.7761	7/9	9	0	no (misses upward)
17	2.0518	**2.0518**	19/20	20	20	**yes**
21	2.1000	2.2589	9/18	18	8	no (misses upward)
31	2.5833	2.7485	11/16	16	3	no (misses upward)
35	2.9167	**2.9167**	15/15	15	15	**yes**
37	3.0345	—	—	2	0	UNSCOREABLE
43	3.0714	—	—	0	0	UNSCOREABLE

Three registered outcomes (Figure 2). **The pooled prediction held**: 46 of 80 valid packings (57.5%) land within 2×10^{-3} of the value predicted by §2's rule, against a registered 30% bar — for a rule fitted entirely on another vendor's models, before this family was ever sampled. The registered metric pools all cells, and the pass rides on that: by §3.2's own convention — arm F's pooled rate excludes non-discriminating cells *because prediction equals the family argmax there* — the decomposition must be stated. 35 of the 46 on-prediction packings sit in the two non-discriminating cells (20/20 at $N = 17$, 15/15 at $N = 35$); on the three discriminating cells the rate is 11 of 43 (25.6%), below the bar the pooled metric passes. The registered verdict stands as registered; what it measures at the discriminating cells is the next finding. **The cell-level prediction did not hold**: the predicted value is modal in 2 of 5 scoreable cells, short of the registered 5, though the registered falsifier (failure in $\geq$4) also did not trigger — and where the family locks on, it locks on almost completely. **The structural signature came in under its bar**: 20 of 46 on-prediction samples

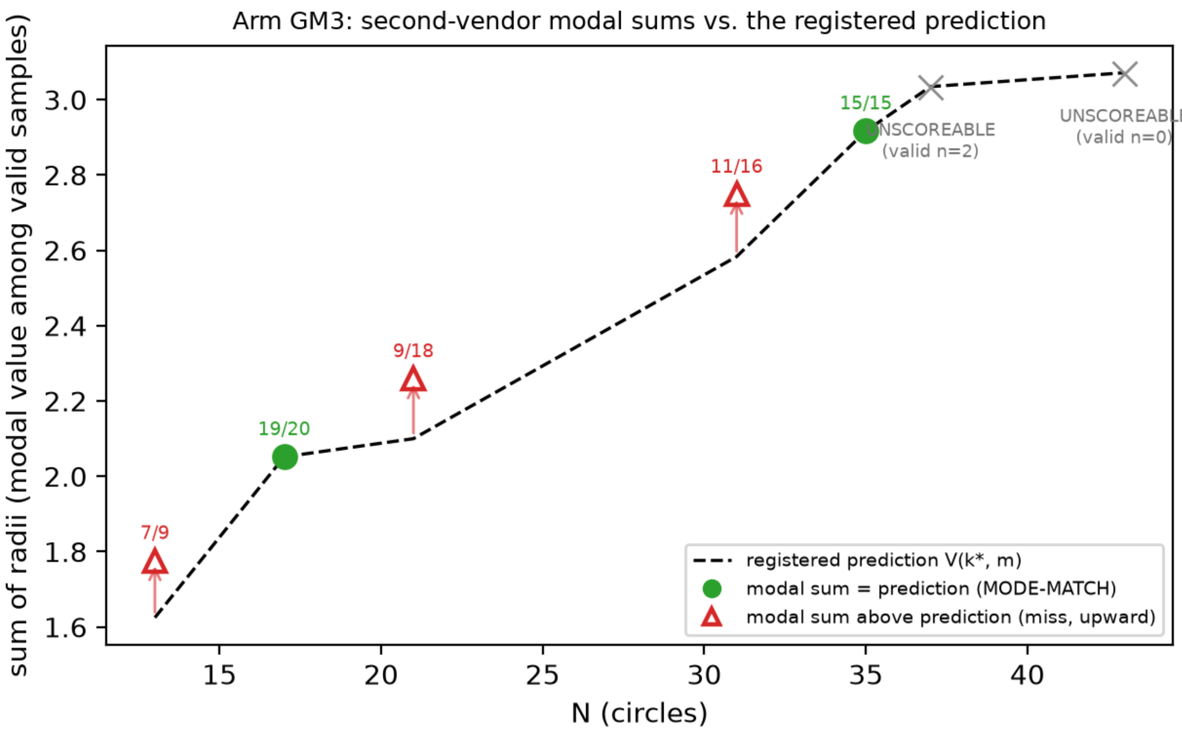

Figure 2: *Arm GM3 modal sums against the registered prediction.* Filled green circles match the predicted mode (frequencies annotated); open red triangles miss — every miss lands above the prediction, on the rival argmax (the modal packings carry the rival's two-radius structure, 27/27 — §3.6); the two largest cells are UNSCOREABLE under the registered <3-valid rule. Regenerates via `fig_gm3_anchoring.py` from `arm_gm3_report.json` (itself replayed from the raw ledger by `arm_gm3_analysis.py`, no arguments).

(43%, bar: half) carry the exact two-radius grid-plus- filler decomposition, so on-prediction Gemma samples reach the predicted *value* through the predicted *construction* slightly less than half the time.

The three cell-level misses are not noise, and "misses upward" understates them: **the family's modal output at every discriminating cell is the rival argmax itself** — 1.7761, 2.2589 and 2.7485 against rivals 1.7761424, 2.2588835 and 2.7485281 — and a post-hoc structural diagnostic (disclosed as such; `diagnostics_gm3_rival.py`, read-only over the ledger) finds all 27 of 27 rival-valued packings carrying the rival's exact two-radius decomposition, the (k^*-1) grid radius with the (k^*-1) filler radius. The registered signature instrument could not see this: it tests radii against the *predicted* order k^*, so it is structurally blind to (k^*-1) constructions — which is why the 43% signature figure above coexists with a 100% rival-structure rate on the miss side. In §2.3's terms, gemma-4-26b performs precisely the drop-to-(k^*-1)-and-fill move that "a proposer with any local search would" perform and that the original family performs in 2 of 57 valid samples: 27 of 43 valid discriminating-cell packings (63%) are the rival construction. Read against §8's tractability alternative, the anchor value marks where construction-by-mental-arithmetic lands, and a family with more arithmetic reach executes the harder in-family construction rather than truncating — same recipe family, other branch. Validity collapsing at the two largest cells (2 then 0 valid of 20) bounds the family's competence window on this task; both cells are UNSCOREABLE under the registered rule (< 3 valid samples, fixed in the GM-chain registration before sampling) and claim nothing.

Arm V. Arm GM3 varied the vendor while holding the instrument close to laboratory grade — first-party API, pinned temperature, explicit output budget, seven cells at twenty samples. Arm V varies vendor and serving class at once, under the same byte-identical arm-F prompts both arms inherit (SHA-256 per N restated in the registration and recomputed at call time, aborting on mismatch; the GM-chain registration pins the identical hashes), at five cells and five samples per (model, N), against free-tier aliases across five vendors (`arm_v_preregistration.md`, committed before sampling; three amendments, each committed before the sampling it governs, expanded the original three-alias design to thirteen aliases across two serving paths — the opencode CLI and OpenRouter chat-completions — and raised `max_tokens` to 24,576 after hidden-reasoning truncation emptied three models' replies). Serving disclosures per the companion paper's protocol: aliases on managed paths, `sampling_params: null` on every row, per-row upstream-provider strings logged where the path reports them, every invocation appended live to `arm_v_candidates_raw.jsonl` — 877 invocations, failures and refusals included. Scoring is `arm_f_repro.py`'s parse/validate/classify unchanged, tolerances registered in arm F before its sampling.

Registered floor: $\geq$10 of a model's 25 registered invocations valid at 10^{-6}; below it, no anchoring claim in either direction. Execution retried each invocation slot until content arrived, per amendment 3's registered transport-repair and accounting rule, so the ledger holds many more attempt rows than slots — gemma-4-31b's 25 slots, for instance, took 190 attempt rows, 165 of them transport failures — and the floor is evaluated on the 25 slots, not the attempts. Ten of thirteen aliases finished BELOW-FLOOR — including all three originally-registered opencode aliases, whose logged failure taxonomies are transport-dominated for one (CLI build banners, timeouts, "Unknown model" errors as the free tier rotated its catalogue) and roughly half transport, half parse-failure for the other two — plus one alias returning all 108 of its attempts without content. Three aliases cleared the floor, and their registered verdicts split:

alias (vendor)	valid	V1 on-prediction	V1 verdict (bar: $\geq$50%)	V2 rival-argmax (bar: $\leq$10%)
`north-mini-code` (Cohere)	12	8/12 [39–86%]	**TRANSFERS** (point est.)	0/4 $\rightarrow$ **HOLDS**
`gpt-oss-20b` (OpenAI, open-weight)	18	11/18 [39–80%]	**TRANSFERS** (point est.)	0/9 $\rightarrow$ **HOLDS**
`gemma-4-31b-it` (Google)	15	0/15 = 0%	**DOES-NOT-TRANSFER**	0/3 $\rightarrow$ **HOLDS**

This is the registration's most informative branch pair. Anchoring under this instrument is not an artifact of the original family — two further vendors' models land the registered construction values at 61–67% of valid outputs (point estimates above the 50% bar; the bracketed Wilson 95% intervals include it, so each verdict is a point-estimate pass at $n = 12$ and $n = 18$) — and the strong form holds everywhere evaluable *within this arm*: zero rival-argmax emissions in 16 discriminating-cell validities (the registration, `arm_v_preregistration.md`, recorded the original family's prior for the same statistic as 0/21 — an undercount against the shipped ledger's 2/23 in §3.3; §3.6's gemma-4-26b, under the GM-chain instrument, is the standing exception — its discriminating-cell mode *is* the rival). The transfer is at reduced concentration: 61–67% on-prediction against 56–76% in-family at the same five cells (§3.2). Reduced concentration at different vendors, on byte-identical prompts, landing on the same modal value is what a shared attractor predicts; equal concentration would be harder to separate from shared training text. And it is not universal: gemma-4-31b, clearing the same floor, lands the prediction never. Its valid packings are unanchored and low (bests 0.88–1.85 against predictions 1.63–3.03) — not the rival, not the template, just weaker de-novo construction. Under the registration's falsification map, the family-specific branch fires — any model clearing the floor with DOES-NOT-TRANSFER makes the closed form a family-bounded property, stated in §5 with data rather than a caveat — while the map's all-transfer branch fails on gemma alone, so the reading that anchoring is an artifact of the original family is defeated by the same table that bounds its generality.

The gemma pair needs stating precisely, because this family now appears on both sides, and the prompts are identical across the two arms — what differs is parameter count, serving path and sample budget, not the elicitation. Under the GM-chain instrument (first-party, pinned temperature, 7×20), gemma-4-**26b** met the pooled bar at 57.5% while its discriminating-cell rate (26%) fell below it (the mixed-serving accounting; first-party-only, 59.0% — §8's registered deviation applies to this figure) while emitting the *rival* construction at 63% of discriminating-cell validities; under arm V's free-tier instrument (5×5), gemma-4-**31b** through a rate-starved alias (Google AI Studio upstream; 27–38 of each cell's attempts returned as shared-pool 429s before content arrived) lands neither the prediction nor the rival — its valid packings are weaker de-novo constructions. The pair bounds the family's behaviour rather than contradicting itself — one member sits *above* the template's execution reach (far enough to build the drop-and-fill rival the original family cannot), the other lands below it — but model variant and instrument quality change together across the two measurements, so which difference carries the split is not attributable from this pair; either way, no single-model measurement would have seen the boundary.

3.7 Arm CC: an executed program does not escape the family

The code-free design (§2.1) rests on a rationale this paper had asserted but not measured: that a code channel routes construction to an executed optimizer, so the emission distribution would reflect the optimizer rather than the model. Deployed discovery loops elicit *programs*, so that gap was the standing objection to this paper's loop relevance. Arm CC measures it. The design was registered and pushed to the public remote before sampling (`arm_cc_preregistration.txt`, commit `285deca`): the three trap cells $N = 13, 21, 31$, $n = 15$ weak-tier invocations per cell, prompt equal to the bare stem with exactly two registered line replacements — the code-free clause becomes a program contract (standard-library `math` only, stated in the prompt) and the output clause asks for program source. Programs run sandboxed (`python -I`, 10-second timeout, registered AST allowlist; the executor and its four-bin smoke test were committed with the registration), and stdout is scored by arm-F conventions unchanged. Two competing predictions, arm-CH style: P-CC1, the modal valid executed output remains the $T(k^*, N)$ bucket at ≥ 2 of 3 cells; P-CC2, it sits strictly above the anchor at ≥ 2 of 3; ties or mixed modes count as neither.

N	Valid	Modal bucket (count, margin)	Anchor-rate	Above anchor
13	12/15	anchor 1.6250000 (2/12, one sample)	2/12	1 — the family argmax, exactly
21	13/15	anchor 2.1000000 (9/13, margin 8)	9/13	0
31	12/15	anchor 2.5833333 (3/12, one sample)	3/12	0

P-CC1 holds as registered (3 of 3 cells), and its margins belong next to it. At $N = 21$ the mode is heavy — nine exact 2.1000000 emissions, most programs writing the 5×5 grid truncated to 21 circles — but at $N = 13$ and $N = 31$ the modal bucket is the anchor value with a **one-sample margin** over dispersed sub-anchor values, so the modal-identity claim is thin at two of three cells, and this sentence is its registered disclosure. The robust result is the registered decomposition row: **no valid program output exceeds the family argmax anywhere** (0 of 37 above the rival), and exactly one exceeds the anchor at all — a single $N = 13$ program that computes the family argmax construction $V(3, 4) = 1.7761424$ *exactly.* That row sharpens arm CH rather than repeating it: handed the family's values, the model chooses the argmax and cannot build it by direct emission (30 of 31 invalid, §3.5); allowed to delegate the arithmetic to an executed program, it reaches the same argmax once in 37 valid outputs and never passes it. Failure taxonomy: 7 of 45 programs produce geometrically invalid packings, 1 imports `random` against the stated contract (caught by the registered gate), none time out. Structural $k = k^*$ in 18 of 37 valid outputs. What the channel changes relative to direct emission is dispersion, not escape: code-free anchor concentration at these cells runs 56–86%, the code channel's pooled anchor-rate is 14/37, and the lost mass sits entirely *below* the anchor — greedy radius-growth heuristics converging under the lattice.

Scope, registered before sampling: this instrument measures externally-executed standard-library arithmetic, not library-assisted optimization — deployed loops permit `numpy/scipy`, and the antecedent study's unconstrained agents wrote `scipy` optimizers unprompted, which is why the constraint is stated in the prompt rather than silently enforced; the library channel remains a further arm. Under the registration's framing map the P-CC1 branch reads: anchoring extends into the math-only code channel at these cells, and §2.1's "a code channel routes around the anchor" is weakened from an assumption to a measured negative — the routing exists, and for this proposer it does not lead out of the family. Raw rows verbatim in `arm_cc_collect.jsonl` (45 of 45 launched invocations collected, no runtime deaths); scoring `arm_cc_analysis.py`; frozen report `arm_cc_report.json` and summary `arm_cc_results.txt`.

Both registered follow-ups hold, and they harden the result. A fresh-draw replication wave (arm CC2) and a Sonnet-tier probe (arm CCS) were registered together (`arm_cc2_preregistration.txt`, commit `4d7b7c5`, pushed before sampling; registered after CC's results were known and disclosed as such), 45 invocations each on the byte-identical prompts. **CC2 returned REPLICATED** under its pre-stated verdict map: 0 of 41 valid outputs above the rival (the registered zero-count prediction, exactly), and the anchor bucket modal at 3 of 3 cells — including $N = 31$, where the mode that was one-sample-thin in CC arrived at 10 of 15 valid with a margin of 8, firming the cell CC could not. Pooled over both weak-tier waves, 0 of 78 valid program outputs exceed the family argmax and exactly one reaches it. **CCS returned P-CCS2**: against competing registered predictions (escape at $\geq 20\%$ above-rival versus a zero-count ceiling), **0 of 37 valid Sonnet-tier outputs exceed the argmax**. The Sonnet programs are qualitatively different search — simulated annealing over hand-rolled linear-congruential generators, Halton-sequence multi-restarts, force-directed relaxation, against the weak tier's greedy radius growth — and the tier barely anchors (5 of 37 on the anchor value, consistent with arm S's direct-emission escape from the template), yet its executed math-only search never passes the family ceiling at any cell. At these cells the ceiling is a property of the channel, not of the tier. Taxonomy: CC2 41 valid, 3 geometry-invalid, 1 blocked import; CCS 37 valid, 7 geometry-invalid, 1 timeout (the programme's first). One protocol note, disclosed: a single CC2 invocation ($N = 13$, slot 2) used runtime tools despite the no-tools dispatch wrapper; per the arm-M convention its verbatim final message is scored like every other row. Ledgers `arm_cc2_collect.jsonl` and `arm_ccs_collect.jsonl` (45/45 each, no runtime deaths); runner `arm_cc2_analysis.py` reuses CC's registered pipeline unmodified. These arms bring the study corpus to 603 invocations.

3.8 Arm CN: the rule predicts the mode where there is nothing to recall

The rule was fitted on $N = 23/26/27$, and $N = 26$ is the canonical published cell. A reader can therefore hold a rival reading of every square-arm result above: the proposer is not constructing a round($\sqrt{N}$) grid but recalling packings it saw, and the closed form happens to describe them. Arm CN separates the two readings by sampling the bare template A.1 at five N that appear in no entry of the sum-of-radii scoreboard cited in §7 and in no prior arm of this study — $N = 50$, 58, 62, 65 and 75, chosen to mirror the square arm's branch mix (three truncation cells, two $m = 1$ extend cells; the $m \geq 4$ branch is already disconfirmed by arm M and was not re-tested). Absence from the pretraining corpus cannot be established and is not claimed; what the arm can establish is whether the rule predicts the mode where no published sum-of-radii packing exists to recall. Two predictions were registered against each other before sampling (`arm_cn_preregistration.txt`, commit `69a1642`, pushed first; motivation disclosed as post-review): P-CN1, construction — the registered value is modal at ≥ 4 of 5 cells including ≥ 2 of the 3 discriminating cells; P-CN2, recall — modal at ≤ 2 of 5. Cells with fewer than five valid outputs were to be reported UNDERPOWERED; ties for the mode count against the prediction. $n = 15$ weak-tier invocations per cell, scored once by the arm-F pipeline unchanged.

N	k^*, prediction	disc.	valid [Wilson 95%]	on-pred.	mode (count, margin)	verdict	k^*-struct.
50	7, $V(7,1) = 3.5295867$	no	11/15 [48–89%]	4	3.5122554 (6, +2)	**MISS**	10/11
58	8, $T(8,58) = 3.6250000$	yes	14/15 [70–99%]	12	3.6250000 (12, +11)	**HIT**	14/14
62	8, $T(8,62) = 3.8750000$	yes	13/15 [62–96%]	13	3.8750000 (13, +13)	**HIT**	13/13
65	8, $V(8,1) = 4.0258883$	no	13/15 [62–96%]	9	4.0258871 (9, +7)	**HIT**	12/13
75	9, $T(9,75) = 4.1666667$	yes	12/15 [55–93%]	8	4.1666667 (8, +6)	**HIT**	8/12

P-CN1 holds. The registered value is modal at 4 of 5 cells, at all three discriminating cells, with margins of 6 to 13 samples; every cell clears the evaluability floor; the family rival is emitted 0 times in 39 discriminating-cell validities and no valid output exceeds the family argmax (0/63). The registered structural secondary holds at 5 of 5 cells: the dominant radius backs out $k = k^*$ in 57 of 63 valid outputs. The one miss is $N = 50$, and its decomposition is the arm-CH pattern: 10 of 11 valid outputs are the registered 7×7 grid plus exactly one filler, 4 with the filler at the interstitial radius $(\sqrt{2} - 1)/14$ (the registered value) and 6 with it pushed into a container corner at $r \approx 0.0123$ — the cell's mode, 0.017 below the prediction. The template is selected; the filler is mis-executed; the value criterion registers a miss, and it is reported as one. At the largest cell, $N = 75$, the four off-mode outputs are alternative constructions (a 5×15 grid, a hexagonal layout, a 12-column layout), none of them the rival $V(8, 11)$. Validity is 63/75 (84%) with no collapse at large N; the invalid rows are nine overlaps, two count errors and one output of fraction literals the registered parser does not evaluate (the CH "radical literal" mode). Raw rows in `arm_cn_collect.jsonl` (75/75 launched, none lost); eleven $N = 65$ rows were transcribed by reconstructing the standard 8×8 grid string from the emitted filler triple, disclosed in `arm_cn_results.txt`; the reconstruction is character-identical for these standard grids, but a reader who discards those rows leaves $N = 65$ with three valid samples, below the five-row floor, and P-CN1 at 3 hits of 4 evaluable cells — the three discriminating cells, which carry the contamination argument, are untouched. The arm brings the study corpus to 678 invocations.

What this does and does not show. The recall reading predicted that the anchor would not survive the move to N with nothing to recall; it survived at four of five cells and structurally at all five. The closed form is therefore a description of what the proposer builds, not of what it has seen, at the branches where it was already supported. This does not establish that the fitted cells are absent from training text, does not extend to the $m \geq 4$ branch, and leaves the canary and perturbed-container probes in §8 unrun.

4 Rectangle transfer

4.1 Transfer of the rule

For a $1 \times a$ rectangle the template has two parameters: $q^* = \text{round}(\sqrt{N/a})$ columns across the width, $p^* = \text{round}(\sqrt{N \cdot a})$ rows across the height. At $a = 1$ both collapse to $\text{round}(\sqrt{N})$ — the same rule with the aspect ratio put back, not a new rule fitted to new data, and no rectangle model output existed when it was written. It was verified against an independent LP at 213 configurations, drift below 10^{-9}, including a cross-domain check: at $a = 1$ with $p = q = k$ the square file's closed form must equal this file's LP. Shape mismatch grows as a leaves 1 — over $N = 10 \ldots 45$ the count of N whose predicted shape differs from the optimal one rises from 12 at $a = 1.0$ to 23 at $a = 1.5$, 20 at $a = 2.0$ and 23 at $a = 3.0$.

We probed the two sharpest cells with sixteen proposers in a container none had been given before: $N = 19$ at $a = 3$ (predicted 3.1666667, rival 3.5749194) and $N = 25$ at $a = 2$ (predicted 3.1250000, rival 3.4832492). With all sixteen scored, **5 of 11 valid proposals landed on the predicted value and 0 of 11 reached the rival**.

We characterize all eleven here. On-prediction: two at $N = 19/a = 3$ (3.1666666667 and 3.16666673, the latter agreeing to six decimals not seven) and three at $N = 25/a = 2$ (3.125 exactly). Off-prediction at $N = 25/a = 2$: two samples emitted a uniform 5×5 grid at $r = 0.1$ summing to 2.5000000 — the *square* rule with no aspect correction, $k = \text{round}(\sqrt{25}) = 5$ — plus one at 3.0 and one at 3.151875. At $N = 19/a = 3$, **two** samples exceeded the prediction from outside the family, 3.45 and 3.5, both below the rival.

We therefore report the rectangle as **partial out-of-sample support, not confirmation**: $5/11 = 45\%$, Wilson 95% CI $[21\%, 72\%]$, and a proposer choosing uniformly among the three or so plausible template shapes would land on-prediction about a third of the time, so 5/11 does not separate cleanly from that null. Two qualifications: validity degrades in the tall container (4/8 at $a = 3$ against 7/8 at $a = 2$, three of four $a = 3$ failures being overlaps), a separate finding and a confound on that cell at $n = 8$; and the rectangle ledger carries no prompt hashes, no run dates and no proposer fields, so the arm with the strongest transfer claim has the least provenance (§9).

4.2 Negative result: the closed form does not survive the move to rectangles

The rectangle generalizes the *rule* but not the *formula*, and we keep the failure because it is what a verification gate is for. In the square every interior vertex has four identical neighbors, so one expression covers every filler. In a $1 \times a$ rectangle, with half-spacings $h_x = 1/(2q)$ and $h_y = a/(2p)$, a filler is also capped by horizontal and vertical spacing to *adjacent fillers*. Those constraints are inactive when $h_x = h_y$ — why the square case could not have revealed them — and bind only when neighboring vertices are occupied, so the cap depends on m and on which vertices a construction uses, and no expression in (p, q, m, a) reproduces it. The LP gate caught this on the first run: at $a = 1$, $p = 2$, $q = 4$, $m = 1$ the natural generalization $\texttt{rf} = \min(\texttt{diag}, h_x, h_y)$ returns **1.125** against a true **1.1545085**, capping against a neighboring filler that does not exist. We retain closed forms only where provably exact (full grid and truncated grid) and use the LP as the value oracle for the extend branch, so what §4.1 tests is the LP-backed prediction.

5 The tier ladder: three attractor families

Holding prompt, container and scoring fixed, we varied only the nominal proposer tier across the three cells that discriminate hardest between prediction and family optimum — $N = 13$, 21 and 31, each inside a trap zone. The tiers do not merely differ in success rate: they attempt qualitatively different constructions.

Table 1: Forecast versus outcome, both containers.

Cell	Branch	Predicted	Rival argmax	Valid / sampled	On-pred / valid	Rival
N=13 (sq)	truncate	1.6250000	1.7761424	4/5 · **18/20**	3/4 · **10/18**	0
N=17 (sq)	extend	2.0517767	†	4/5	3/4	†
N=21 (sq)	truncate	2.1000000	2.2588835	7/10 · **15/20**	4/7 · **12/15**	2
N=31 (sq)	truncate	2.5833333	2.7485281	5/5 · **17/20**	5/5 · **13/17**	0
N=35 (sq)	truncate	2.9166667	†	4/5	3/4	†
N=37 (sq)	extend	3.0345178	†	4/5	3/4	†
N=43 (sq)	truncate	3.0714286	3.2416246	7/10	6/7	0
N=19, a=3	truncate	3.1666667	3.5749194	4/8	2/4	0
N=25, a=2	truncate	3.1250000	3.4832492	7/8	3/7	0

Plain entries are the original 45-invocation bare arm (§3.3's corpus); **bold** the same cells over the full 85-row bare ledger after §6 added 40 rows. † marks non-discriminating cells (prediction equals family argmax), excluded from rival denominators. Square discriminating cells: 18/23 and 2/23 on the original corpus; 41/57 and 2/57 on the full ledger. Rectangle: 5/11 and 0/11. Sources: `arm_f_repro.py` over `arm_f_candidates_v2.jsonl`, `arm_g_rect.py` over `arm_g_candidates.jsonl`.

The weak tier truncates templates. The Haiku-tier proposer (Figure 3, left panel) produces uniform grids of radius $1/(2k)$ truncated to N circles, with corner fillers only when the grid underfills. Across the 45 original bare invocations spanning $N \in \{13, 17, 21, 31, 35, 37, 43\}$, **35 were geometrically valid at the primary 10^{-6} tolerance (77.8%), 29 at 10^{-9} (64.4%)** — recomputed from the ledger, superseding the figure carried in earlier drafts. At the three discriminating cells 12 of 16 valid samples landed on the predicted value and the rival was reached twice, both at $N = 21$.

The middle tier perturbs and mixes. The Sonnet-tier arm was preregistered in `arm_s_preregistration.txt`, disclosing that 5 of 20 samples at $N = 13/21$ had been seen before registration and that $N = 31$ was fully blind. It was valid 30/30 at 10^{-6} and 27/30 (90%) at 10^{-9} — so the "100%" leg is tolerance-dependent and both numbers are stated. It was almost never on-prediction: 0/10, 0/10, 1/10 — 1/30 pooled against the weak tier's 12/16 at the same cells. Instead of truncating a uniform grid it perturbs one, with enlarged edge rows, hexagonal interior rows and two or three distinct radii in the same packing (29/30 multi-radius).

One Sonnet sample at $N = 31$ emitted 27 circles at $r = 1/12$ with 4 corner circles at $r = 1/8$, summing to 2.7499999991 — above 2.7485281, the best the family reaches at that N. Recomputed from its stored coordinates, the construction is tangency-tight: minimum pairwise and wall slack both exactly zero at tolerance zero, the exactly-tangent contacts involving the binary-exact $r = 1/8$ corner circles while the $r = 1/12$ circles carry small positive slack. It is the only sample in the study to leave the recipe family upward.

That sample also exposes a scoring defect. At $N = 31$ the family rival (2.7485281) and this out-of-family value (2.75) differ by 1.47×10^{-3}, *inside* the registered 2×10^{-3} window, so the value classifier counted the escape as a hit on the construction it exceeded. The window was chosen for the anchor comparison, where the nearest competing value is ∼0.15 away, and reused where it does not fit. Classified by structure instead, Sonnet's rival count at $N = 31$ is 2/10 and pooled 5/30. Every categorical claim in this paper about "reached the rival" or "left the family" is made at structure level or at 10^{-6}, never at 2×10^{-3}.

The top tier attempts recursive constructions and fails to build them. The third arm was invoked through a bare tier alias. We name it `opus_alias` throughout and make no claim about which weights served it: the runtime accepts only the bare alias and exposes no dated identifier, so the binding is a vendor promise rather than an attestable fact. No version number appears anywhere in this paper.

That caveat matters: completion-time and token-count anomalies were observed in the unreleased session transcript for this arm, but the ledger carries no latency or token field, so they appear in no released artifact and nothing below relies on them.

Table 2: Three-attractor ladder. Valid columns count samples passing containment and non-overlap at the 10^{-6} and 10^{-9} tolerances; On-pred and Rival count valid samples landing on the predicted value and on the rival argmax.

Tier	Cells	n	Attempted family	Valid 10^{-6}	Valid 10^{-9}	On-pred	Rival	Failure mode
haiku (bare)	13, 17, 21, 31, 35, 37, 43	45	uniform grid, truncated; fillers when underfull	35 (78%)	29 (64%)	12/16 (3 matched cells)	2	overlap 6, outside 1, parse 3
haiku, matched cells	13, 21, 31	60	as above	50 (83%)	—	35/50	2	overlap 5, outside 2, parse 3
sonnet	13, 21, 31	30	perturbed grid, hexagonal rows, 2–3 radii	30 (100%)	27 (90%)	1/30	5 (structure)	none
`opus_alias` ‡	13, 21, 31	30	quarter-circle corners, Apollonius fillers, coarse grid + border strips	4 (13%)	4 (13%)	0/4	0	overlap 24, non-positive radius 2

Under those caveats the arm — preregistered fully blind in `arm_o_preregistration.txt` — was valid in 4 of 30 invocations (13%) at both tolerances. At every cell it attempted a more ambitious construction than either other tier: quarter-circle corners at $r = 0.25$ with Apollonius-style fillers at $N = 13$ (10/10 samples), mixed-radius 4×4-ish grids at $N = 21$, and a coarse 3×3 grid at $r \approx 1/6$ with border strips at $N = 31$ (0/10 valid). Failures are geometric rather than numerical — edge strips at $r = 0.03$ placed 0.138 from an $r = 1/6$ grid circle needing 0.197 — and twice the arm padded to count with zero-radius circles. Valid samples score *below* the trap they were expected to fall into (1.26–1.41 at $N = 13$ against 1.625). A post-hoc diagnostic (not preregistered; labeled as such in the repository) finds no evidence of a tolerance artifact behind the 13% in this sample: 24 of 26 invalid samples overlap grossly (median maximum overlap 3.3×10^{-2}; radius-shrink repair costs a median 15% of sum-of-radii), while tolerance-scale near-misses ($< 2.5 \times 10^{-5}$) occur instead in 5 of 7 geometry-scored weak-tier failures — the exact-tangency grids, not the ambitious constructions.

Two decisions here were made after seeing the data and are recorded as deviations (Table 4). P-O1, P-O2 and P-O4 are reported **not evaluable**, on the grounds that a validity collapse of this size makes an on-prediction tier comparison uninterpretable; P-O3 is trivially satisfied on 4/4 valid samples; the registered disconfirmation — regression toward the trap — did not occur. De-registering three of four predictions in the one arm that disconfirmed is exactly the move preregistration exists to constrain, which is why it is tabled rather than narrated. And the arm was initially excluded from the ladder, then included once its results were known; we follow the later decision and report the ladder both ways (Table 2) — with the arm 83% → 100% → 13% on matched cells (78% over all seven weak-tier cells), without it 83% → 100% plus the qualitative shift from truncation to perturbed multi-radius hybrids, which is already the publishable contrast.

‡ Addressed through a bare serving alias; the alias-to-weights binding is not attestable, and this row is not a statement about any released model version. Tier denominators are not matched (the weak tier spans seven N against three), which is why the matched-cell row is given separately, and it is the matched comparison (83% → 100% → 13%) that should be quoted across tiers. Sources: `arm_f_candidates_v2.jsonl`, the three preregistration files.

"Monotone" applies to one axis only: ambition rises monotonically — truncated template, perturbed hybrid, recursive gasket — while validity rises then collapses. "Ambition" is measurable, not merely a qualitative read: a post-hoc diagnostic (disclosed; `diagnostics_ambition.py`, all samples with emitted geometry, invalid included) gives median distinct radii per sample 1 → 2 → 3 (means 1.33 → 2.40 → 3.37) and median off-lattice center fraction 0.08 → 0.43 → 0.95 across the three tiers — monotone on both operationalizations. At the canonical and plausibly contaminated cell $N = 26$ all tiers converge on the same 2.5414 attractor; they diverge only at withheld trap cells. The branch rule of §2 is therefore a weak-tier regularity — cross-vendor under two instruments (a registered pooled bar under GM3's, carried by its non-discriminating cells, §3.6; 61–67%

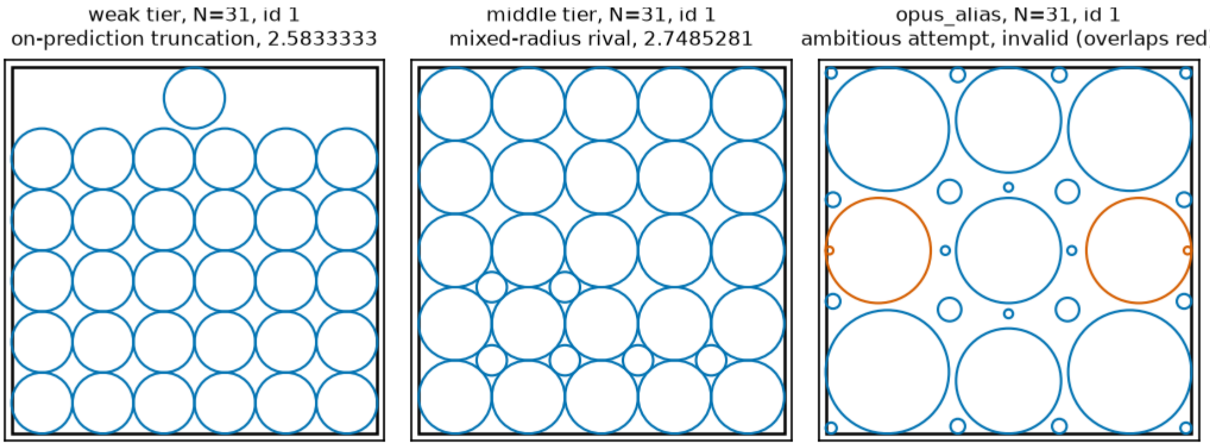

Figure 3: *Three attractor families at a single cell, $N = 31$.* One representative packing per tier, ledger sample id in each panel title (weak tier `bare` id 1, on-prediction truncation 2.5833333; middle tier `sonnet_bare` id 1, mixed-radius rival 2.7485281; `opus_alias` id 1, invalid, with only the offending circles highlighted). Source: `arm_f_candidates_v2.jsonl` via `fig_scripts.py`, deterministic (the exemplar rows' geometry is byte-identical in v1; v2 differs only in provenance metadata).

of valid outputs in two further vendors under arm V's, §3.6), family-bounded in both directions (absent in gemma-4-31b at the same floor, escaped *upward* to the family rival by gemma-4-26b, §3.6), cell-level modes confirmed in one lineage — and this is a **boundary condition on the main result**, not an independent second finding. It also cuts against §1's framing: the discovery systems cited there run frontier or mid-tier proposers, so the tier at which our regularity holds is the tier they do not use, while the tier closest to what they use escapes the trap in our own data.

6 Elicitation and faithfulness

This section is explicitly secondary material and the result leans negative: the registered validity prediction fails, the concentration effect survives only as a fragile directional finding, and the faithfulness audit checks description against artifact, not against process.

6.1 Design and bundling disclosure

A ten-versus-ten pilot at $N = 21$ compared the bare prompt against a variant asking the proposer to name its construction on a leading `METHOD:` line: validity rose from 7/10 to 10/10 and the higher-scoring rival disappeared, from 2 of 7 valid bare samples to 0 of 10 trace samples. Before any scaled sample was drawn we registered four predictions and an explicit falsifier in `arm_t_preregistration.txt` (sha256 `ab7900a8...`), disclosing that the pilot's trace prompt had drifted from the bare template beyond the method line — it omitted the `[0,1]x[0,1]` tokens and reworded the output-format line — so the pilot's intervention was method-line-plus-rewording, bundled. Pilot samples are never pooled with `trace_v2`.

The scaled prompt, `trace_v2`, is itself a **bundled prompt-format-and-trace-request** intervention, though a near-minimal diff: one inserted `METHOD:` line, plus `"After the METHOD line, "` prepended to the output line and `"no other text"` changed to `"no other text after the list"`. That second change is not cosmetic — §6.2 shows it doing measurable work — so the measured effect is *method-line-plus-output-line-rewording*, not the method line alone: a weaker version of the confound for which the pilot was discarded, held to the same standard.

The scaled arm ran 100 new invocations, bringing both arms to 20 per cell at $N \in \{13, 21, 31\}$ and the corpus to 215 logged square invocations (85 bare = 45 original + 40 new; 70 trace = 10 pilot + 60 trace_v2; 30

Table 3: Paired trace grid (scaled arm only).

N	Arm	n	Valid (parse / geom. fail)	On-pred / valid	Rival
13	bare	20	18 (0 / 2)	10 (56%)	0
13	trace_v2	20	18 (0 / 2)	16 (89%)	0
21	bare	20	15 (1 / 4)	12 (80%)	2
21	trace_v2	20	18 (0 / 2)	16 (89%)	0
31	bare	20	17 (2 / 1)	13 (76%)	0
31	trace_v2	20	17 (0 / 3)	14 (82%)	1
pooled	bare	60	50 (3 / 7)	35 (70%)	2
pooled	trace_v2	60	53 (0 / 7)	46 (87%)	1

sonnet; 30 `opus_alias`), plus 16 rectangle invocations, 231 total before the extension arms; arm M (§3.4) added 57 sampled weak-tier invocations (60 launched), bringing the study corpus to 288, and arms MU and CH (§3.5) added 180 more (180 launched, none lost), and the code-channel arms CC, CC2 and CCS (§3.7) added 135 more (135 launched, none lost), bringing the unconditioned-and-elicitation corpus to 603, and the held-out-N probe CN (§3.8) 75 more, to 678; the cross-vendor chain is scored under its own registrations and adds 420 GM-chain invocations (§3.6, §8) and 877 arm-V attempt rows over 325 registered invocation slots (§3.6). One disclosure: **20 of the 60 bare samples are pre-existing arm-F rows** ($N = 13$ ids 1–5, $N = 21$ ids 1–10, $N = 31$ ids 1–5) whose results were known when P-T1–P-T3 were written; only trace_v2 is fully fresh (§6.4). Scoring used `arm_f_repro.py` unchanged with the registered 2×10^{-3} window; the Fisher exact test comes from the hypergeometric tail in `arm_t_analysis.py`, checked against a reference.

6.2 Registered outcomes

Preregistered criteria are evaluated exactly as registered, and unregistered alpha thresholds are applied nowhere.

P-T1, validity: NOT confirmed. P-T1 is the one prediction that registered an alpha ("validity $\geq$ bare at EACH N, and pooled one-sided Fisher $p < 0.05$"). Direction held at all three cells, but the pooled test gives $p = 0.30$: 53/60 (88%) trace_v2 against 50/60 (83%) bare, a difference whose detection needs several hundred samples per arm at 80% power — a failure to detect, not a demonstration of no effect. The direction that did hold is not geometric: bare had **3 parse failures and 7 geometric failures**, trace_v2 **0 parse and 7 geometric**, so among parsed samples geometric validity is 50/57 (87.7%) versus 53/60 (88.3%). The P-T1 direction is format compliance, exactly what the reworded output-format line manipulates; validity is reported split by cause from here on.

P-T2, rival suppression: CONFIRMED as registered (directional, no alpha): 1 of 53 valid trace_v2 against 2 of 50 valid bare. Counts this near zero carry no inferential weight and we claim none (Fisher $p = 0.48$, carrying no decision). One trace_v2 sample at $N = 31$ hit the rival 2.7485281 to within 3.6×10^{-8} — the first weak-tier rival hit at $N = 31$ in any arm, and formerly one of the scorer's mismatches (§6.5).

P-T3, anchor concentration: met as registered (directional), inferentially fragile. Among valid samples, 46 of 53 trace_v2 (87%) landed on the registered prediction against 35 of 50 (70%) bare. The p-value it never carried is 0.0325, one-sided uncorrected (§6.4).

P-T4, faithfulness: confirmed under a corrected scorer. The registered scorer gives 38 of 41 scoreable claims matching (93%) against a 90% threshold; blind hand-adjudication under a rubric frozen before rescoring gives 54 of 56 (96.4%). §6.5 reports both.

Per-cell one-sided Fisher on on-prediction (Figure 4): $N = 13$ $p = 0.030$; $N = 21$ $p = 0.41$; $N = 31$ $p = 0.50$. Pooled $p = 0.0325$; Holm threshold over the registered family ($m = 3$ tested with p-values) is 0.0167. The pilot (10 invocations at $N = 21$, 10/10 valid, 9/10 on-prediction) is reported separately and never summed into any total. Regenerated by `arm_t_analysis.py` over `arm_f_candidates_v2.jsonl`.

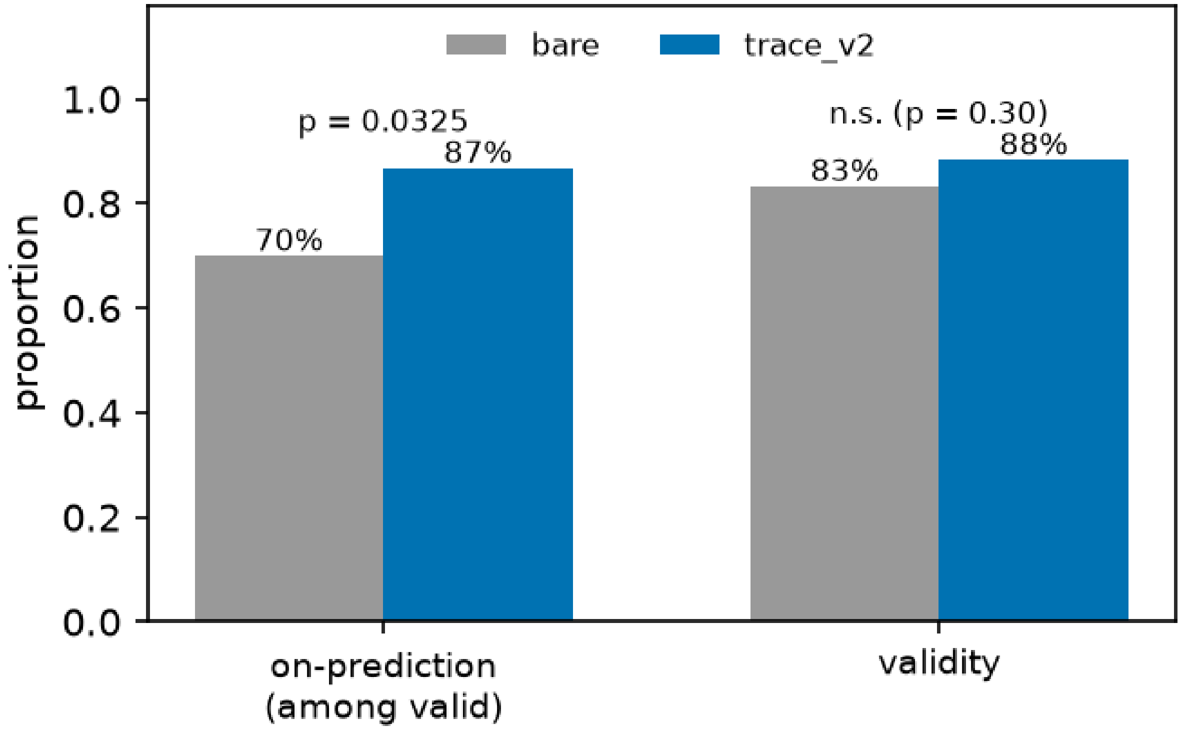

Figure 4: *Bare versus `trace_v2`, by cell and pooled.* Caption caveat: the bars pool the bare arm's two collection waves, which §6.4 shows drift materially, so the pooled bare bar is a mixture; the same-wave comparison in §6.4 is the wave-controlled picture.

6.3 The registered falsifier, both readings

The registered falsifier reads: *"if trace_v2 validity `<=` bare at 2+ of 3 N, the pilot effect was the bundled rewording, not the method-line request — reported as such, not reframed."* Observed validity: $N = 13$, 18/20 both arms — **tie**; $N = 21$, trace_v2 18/20 against bare 15/20 — trace higher; $N = 31$, 17/20 both arms — **tie**. A tie satisfies `<=`, so under the registered wording the falsifier is **triggered at 2 of 3** N, and we report it as triggered; the code instead evaluated direction as `t_rate >= b_rate`, strict `<`, under which it fires at 0 of 3 cells. That convention appears nowhere in the preregistration, was chosen at analysis time, and read the same tie favorably twice — as "direction held" for P-T1 and as "not a falsifier cell" here.

We name the failure mode **operationalization drift between preregistration text and analysis code**: registered prose leaves boundary behavior — ties, inclusive versus exclusive bounds, rounding — implicit, and the scorer must choose after the data exist; the preregistration literature (§7) discusses what is locked, not this tie-handling layer. Remedies, free before sampling and impossible afterward: register the comparison as executable code, or state the tie convention in the registration text. `arm_t_analysis.py` now prints both readings and names the registered tie-inclusive one authoritative. Under that reading **the pilot's validity effect is attributed to the bundled rewording, not to the method-line request**, as the falsifier clause instructed; the §6.2 parse-versus-geometric split points the same way, and P-T1 fails under both readings regardless.

6.4 Limits of P-T3: what the concentration result will and will not carry

P-T3 met its registered directional criterion. It is not a flagship result.

No registered alpha. P-T3 registered direction only; the p-value is a post-hoc addition.

Multiplicity. Of four registered predictions, three are reported with p-values. Holm sets the first threshold at $0.05/3 = 0.0167$; $p = 0.0325$ fails it, as it fails Bonferroni at any $m \geq 2$ and Benjamini–Hochberg FDR at $q = 0.05$. Two-sided Fisher is $p = 0.0537$. P-T3 is nominally significant uncorrected and is not under family-wise error control.

Concentration in one cell. Per-cell tests give $p = 0.030$, 0.41, 0.50 (Table 3); the pooled 17-point gap is a 33-point gap at $N = 13$ averaged with 9- and 6-point gaps elsewhere, driven by a *low bare rate at* $N = 13$ ($10/18 = 56\%$) rather than a high trace rate — the trace rate there (89%) equals that at $N = 21$. Dropping $N = 13$, the comparison is 30/35 vs 25/32, $p = 0.31$.

Wave confound. The bare arm mixes two collection waves; trace_v2 is one. Splitting bare on the registered id boundary, with no intervention applied to either half, on-prediction is: $N = 13$, 3/4 old wave vs 7/14 new; $N = 21$, 4/7 vs 8/8; $N = 31$, 5/5 vs 8/12 — the control arm's own between-wave drift is comparable to the effect attributed to the manipulation. (An earlier draft printed 2/4 vs 10/11 at $N = 21$ and a 25/37 pooled figure — arithmetically impossible against §6.1's id boundary; corrected here from the ledger.) Restricting to one wave, new-wave bare is 23/34 (68%) against trace_v2's 46/53 (87%), but **this stratified comparison is post hoc, not preregistered, no confirmatory weight** — a diagnosis, not a rescue: wave is a confound *in the registered analysis*, which cannot separate it from the manipulation. The design fix, interleaved collection in one session, was not run.

What we therefore claim. A minimal method-naming request does not make the proposer better at geometry and does not measurably change which rare constructions it reaches; in the registered direction it concentrates output onto the anchor, but that concentration fails correction, is carried by one cell, is confounded with collection wave, and is bundled with an output-format rewording that demonstrably affects parse compliance. Studies collecting process descriptors *by asking for them*, including descriptor-driven quality-diversity pipelines where the descriptor is a requested self-report, should treat this as a reason to check, not a coefficient to apply. The pilot also showed a *third* effect in the P-T3 direction at the same cell (trace 9/10 on-prediction against bare $N = 21$ at 4/7, one-sided $p = 0.16$), so P-T3 replicated a pilot signal rather than testing a blind prediction — a different status, and the prereg's pilot disclosure covered the prompt drift only.

6.5 Faithfulness with ground truth

Claim and artifact sit in the same completion and the artifact is fully determined, so a method line is checkable against the emitted coordinates. `arm_f_repro.trace_faithfulness()` matches numeric dimensions from the method line against the emitted layout signature, returning 38 of 41 scoreable claims matching (93%) against the registered 90% threshold. Its 12 exclusions are regex coverage gaps, not claims without numeric content: `DIMS_RE` matched only the literal `NxN` form and `ROWSCOLS_RE` required rows before columns, so these fell through despite carrying explicit checkable dimensions — "Regular 3 by 4 grid with one additional circle in row 4"; "3 complete rows of 4 circles and 1 centered circle in the top row"; "rectangular grid 4-4-4-1 with uniform radius 1/8"; "Four-row grid with 4 circles in the first row and 3 in each of the remaining"; "Rectangular grid arrangement with 5 columns and 4 rows, plus one additional circle on top edge"; "Square grid 5 columns 4 rows with 1 additional circle at top"; "Rectangular grid six columns by six rows radius one-twelfth with five circles removed"; "Regular 5-column by 6-row grid plus one additional circle in the right margin"; "Rectangular grid packing with five columns and six rows plus one additional circle". Three further exclusions — "Uniform horizontal strips with tight row spacing", "Regular rectangular grid packing with corner circle", "Equal-radius rectangular grid with 4 rows stacked vertically" — are closer to genuinely unscoreable. The excluded set is enriched for non-canonical phrasings, where a mismatch is a priori *more* likely, so the exclusion was not conservative in a known direction; worst case, had all 12 been unfaithful, the rate would have been $38/53 = 72\%$.

Blind rescore under a frozen rubric. The rubric (`p7_faithfulness_rubric.md`) — checkable assertions in any surface form, plus match rules including a base-grid convention for "grid plus additions" claims and a truncation convention for "with k removed" claims — was committed to git as `e181d2a` at 03:56:56 on 2026-08-01, **before** any rescoring run, so the freeze is checkable, and it pre-committed the reporting rule: below 90%, P-T4 reported not confirmed, full stop, no consolation framing. Scoring was blind — claim text and circles only, no predicted or rival values, no running tally, no threshold and no sight of the existing 38/41 result — with tallying afterward; all 60 labels are released verbatim in `p7_blind_labels.json`.

Result. Over all 60 trace_v2 rows: **54 MATCH, 2 MISMATCH, 4 UNSCOREABLE** — a corrected rate of **54/56 = 96.4%** of scoreable claims, Wilson 95% CI [88%, 99%], passed, and on 56 scoreable claims rather than 41. The CI's lower bound sits just below the registered threshold, so "clears 90%" is a statement about the point estimate at this n. The two mismatches share a form: row 11 claims "alternating rows of 4 and 3 circles" over observed row sizes 4,3,3,3, row 33 claims "5 rows alternating between 5 and 4 circles" over observed 5,4,4,4,4 — both *mis-descriptions of alternation*, a repeating pattern named and built for the first period only. The original scorer's three mismatches were three different claim forms, not one; the filler-adds-rows mechanism the original scorer described was real, and the rubric's base-grid convention handles it, so all three score MATCH under hand adjudication.

Scope, three ways. The blind pass covers *invalid* rows too: the original 93% conditioned on completions that produced a correct packing — the wrong conditioning, since a method line attached to an overlapping layout is where description and artifact are likeliest to diverge — while the blind pass scored all 60 rows including the 7 invalid ones, 4 of the 60 unscoreable. Second, the check verifies the description against the *artifact*, not the process: the non-claim guard applies in full, and for a model already emitting grids, "5 × 5 grid" matching a 5×5 grid is closer to a self-consistency check than to the causal question the chain-of-thought literature asks (§7). Third, the audit is confounded with the elicitation arm, computed on the arm §6.2 shows concentrates 87% of valid outputs onto one template, where correct description is close to free; faithfulness is informative precisely on off-template outputs, such as the pilot's triangular-hexagonal 6+5+4+3+2+1 sample, described exactly while summing to 1.75. The split by on- versus off-prediction is not run here (§9, stopping rule). What survives is narrow: in a domain where claims are checkable, the method lines this proposer emits are, at 54 of 56 scoreable claims, true of the object it produced — and both failures are a model describing a pattern it started and did not finish.

7 Related work

LLM-driven discovery and the circle-packing scoreboard. Language Model Crossover (Meyerson et al., 2024), ELM (Lehman et al., 2022), EvoPrompt (Guo et al., 2024), EoH (Liu et al., 2024) and LLaMEA (van Stein & Bäck, 2025) established the LLM call as an EC operator over prompts and programs; FunSearch turned the pattern into a discovery claim, and every system in §1's scoreboard inherits it. Two orderings in that scoreboard matter: HELIX's figure is marginally *below* ShinkaEvolve's while being described as state of the art, and ThetaEvolve claims new best-known bounds, so the benchmark is saturated only across the AlphaEvolve–ShinkaEvolve–HELIX cluster and the number is still moving. OpenEvolve, the open-source AlphaEvolve reproduction most practitioners run, reports ≈2.635977. §1.1 records the correction to our record comparison and its $N > 30$ coverage gap. Two critiques bear on our framing and are frequently conflated, so we attribute them explicitly: Gideoni et al. (2026) show simple baselines recover much of the reported advantage, and Berthold et al. (2026) show classical solvers do too.

Template convergence and diversity collapse. "Mutation Without Variation" (Gurkan et al., 2026) finds iterated LLM program mutation collapsing onto previously seen structural templates — in 87% of chains, over 93% of mutations revisit a previously seen form — which is the same bias family in a different regime: mutation loops, program space, no closed form, no preregistration. On anchoring, Huang et al. (2026) treats anchoring as a general bias over initial information while Lou & Sun (2024) and Malberg et al. (2025) document numeric primes dragging point estimates; our modality is the difference, since a construction template has a computable value where a scalar anchor has only a measurable pull. Concurrently, "Measuring the Gap Between Human and LLM Research Ideas" (Chen et al., 2026) reports the same collapse one level up, in idea space: across 11,683 paired proposals, LLM research ideas concentrate on bridge-and-synthesis framings (47–64% versus 12% for humans; normalized entropy down to 0.55 versus >0.92), and chain-of-thought pushes the distribution *further* from the human one. That result is distributional; ours is the limiting case where the collapsed distribution's mode is a single closed-form-predictable object — narrowness measurable there, computable here. "Artificial Hivemind" (Jiang et al., 2025) documents the same homogeneity across open-ended domains at survey scale, which makes cross-domain generality of the collapse the expected default;

against that backdrop our contribution is not that collapse occurs but that, in this regime, its mode is a single closed-form-computable object, stated before sampling. Converging skepticism about what the proposer contributes comes from "Dictionaries, Not Darwin" (Li, 2026), Wang et al. (2026), BehaveSim (Zhang & Lu, 2026), Strategy Diversity (Yang et al., 2026), the bin-packing critiques (Herrmann & Pallez, 2026; Sim et al., 2025) and MathConstruct (Balunović et al., 2025). Two results cut the other way and belong on the other side of the ledger: Zhang et al. (2024) provides empirical grounding for the *importance* of evolutionary search, which is evidence against our substitution of an unconditioned call for the loop and is engaged in §8; and Zhang et al. (2026b) finds strong LLM optimizers act as *local refiners*, which is what a parent-conditioned call would be, so it bears on §8's mutation arm.

Scaling inversions. Zhou et al. (2024) show larger and more instructable models becoming less reliable; the o3/o4-mini system card shows the same on PersonQA; Shojaee et al. (2025) reports collapse past a complexity threshold; GeoBuildBench (Kim et al., 2026) finds geometric construction a regime where nominal capability and executed correctness diverge. Our instantiation is *attractor family* versus executability.

Trace faithfulness, and what our result is not. One cluster intervenes *on trace content* and finds answers do not move — Reasoning Theater (Boppana et al., 2026), Project Ariadne (Khanzadeh, 2026), "Beyond the Commitment Boundary" (Scalena et al., 2026) and "The Chain Holds, the Answer Folds" (Li et al., 2026) all report causal decoupling. A second cluster *estimates* faithfulness without observing the process, represented here by Arcuschin et al. (2026); §6.5 reports the case where ground truth for the described artifact exists. Our result sits at a third intervention point: we vary whether a trace is *requested* rather than editing its content, which is measurement reactivity, not causal faithfulness. The closest cousin is "The Price of Format" (Yun et al., 2025), showing format constraints collapse generation diversity — and as §6.1 concedes, our manipulation includes a small output-format change, so we commit a miniature version of the confound we cite. Format-restriction performance costs are documented at scale in Tam et al. (2024). Related: (Abdelnabi & Salem, 2025; Zhang et al., 2026a; Wu et al., 2025).

Preregistration lineage. Thomas et al. (2026), Williams et al. (2026), Vaccaro (2026) and Zhang (2026) preregister recipes, outcome-blinded predictions and agent protocols; HindsightBench (Jia, 2026) releases frozen preregistrations of directional aggregate hypotheses. We adopt the standard rather than claim it. What we add is the object locked — exact closed-form point predictions of specific output values — and, from §6.3, the observation that locked prose and executing scorer can disagree at the boundary. Structural precedents: Kaplan et al. (2020) and Hoffmann et al. (2022) publish a functional form then run the confirming instance; Snell et al. (2024) is the closest structural twin. QDAIF (Bradley et al., 2023) is the QD-descriptor setting most affected by our §6 result, and GigaEvo (Khrulkov et al., 2025) the concrete in-scope system running MAP-Elites with LLM-driven mutation, where a self-reported behavioral descriptor would be perturbed in practice.

8 Limitations and untested alternatives

The measured distribution is not the in-loop distribution. We sample generation-0, parent-free, code-free, feedback-free calls; the systems in §7 sample parent-conditioned program proposals under selection. The claim that survives is about unconditioned calls, and Zhang et al. (2024) is published evidence that evolutionary search contributes materially, cutting against any stronger reading. The decisive cheap experiment is a one-parent mutation arm: same cells, prompt = bare template plus one parent packing plus its score plus "propose a modification that increases the sum of radii", under three parent conditions (the on-prediction truncated grid, the higher-value family rival, an off-family parent), measuring whether the branch rule still predicts and whether the model inherits the parent's k. That experiment has now been run as the preregistered arm MU (§3.5), and its registered falsifier triggered: 0 of 26 valid rival-parent samples return to the anchor, keep-or-improve and k-inheritance are both 26/26, and the anchoring result is scoped to unconditioned calls as a stated negative transfer result. The concern was correct.

A mechanism-free alternative, now directly probed. Under an explicit "construct the packing by reasoning alone" constraint, a $k \times k$ grid at $r = 1/(2k)$ is close to the only construction whose coordinates can be emitted from mental arithmetic without error accumulation, and round($\sqrt{N}$) is the arithmetically nearest such grid. That hypothesis predicts every observation in §2–§3 without appeal to stored templates, and predicts §5's inversion too: the more ambitious tier attempts constructions whose coordinates it cannot compute by hand and overlaps — precisely the observed failure mode (24 overlaps and 2 nonpositive radii out of 26 `opus_alias` failures). The separating experiment — hand the model the recipe family explicitly, state $V(k, m)$ and the admissible k, and ask it to *choose* k; if it picks the argmax, the tractability reading stands — has now been run as the preregistered arm CH (§3.5). The outcome splits the hypothesis in two: the model's *choice* follows the stated score table (so tractability does not explain the selection), but its *execution* fails off-template in 30 of the 31 invalid attempts (so tractability does explain the instantiation — the grid is what survives mental arithmetic). §1's vocabulary remains behavioral and no memorization claim is made. A second mechanism-level account, typicality bias in preference-tuned models (Zhang et al., 2026a), predicts concentration on the most typical construction without reference to geometry; it is not excluded by any observation here and was not tested against the tractability reading.

Contamination is probed at one level only. $N = 26$ is the canonical published cell and plausibly in training data; membership and extraction probes of the kind used to detect benchmark contamination (Carlini et al., 2021; Golchin & Surdeanu, 2024) were not run. Arm CN (§3.8) answers the comparison against N values absent from the literature: at five held-out N the registered value is modal at four cells and the template structure at all five, against a registered recall alternative. What remains unrun is a canary-string test and a perturbed-container test preserving difficulty while destroying lexical overlap; neither can be run on the present ledgers, and absence of the fitted cells from the pretraining corpus is not established by any of this.

Vendor scope, and what the cross-vendor chain cost to learn. All tiers in the main study come from one provider; §3.6 is the arm that moves the pooled-level claim across vendors. The instrument's history matters as much as its verdict and is recorded here. Arm GM (gemini-2.5-flash-lite, prereg commit `37b3adb`) stalled on first-party quota at 55 of 140 and was completed through a routed serving path under a registered amendment (`arm_gm_amendment1_openrouter.md`, provider logged per row — every resumed row reports Google as the upstream). Its registered verdict is UNDERPOWERED in both accountings: 20 valid packings of 140, three scoreable cells against the five the primary requires, pooled on-prediction 15% against the 30% bar, falsifier not triggered (`arm_gm_analysis_v2.py`, `arm_gm_v2_report.json`). Two descriptives, claimed as nothing more: the one mid-size scoreable cell matches the predicted mode exactly (2.1000 at $N = 21$, 3/6), and off-prediction valid sums land far *below* the prediction (0.48–1.55) — the opposite direction from GM3's misses, which land on the family rival (§3.6), consistent with §5's point that the regularity is tier-bound rather than universal. Arm GM2 (gemma-4-26b-a4b-it, prereg `3019aab`) completed 140/140 invocations with 0 of 140 parseable — and the two candidate readings (format inability versus budget truncation) were separated by GM3 as registered: the vendor API returns deliberation as a separate thought channel, GM2's 4,096-token budget died entirely inside that channel, and at 16,384 tokens the same model parses at 118/140. A cross-vendor null under a fixed output budget is not evidence about the model until the budget's interaction with the vendor's response structure is itself checked. One mechanical deviation on GM3 is on record (amendment registered before resumption): the final 41 of 140 cells were collected through a routed third-party serving path after the first-party quota stalled, provider logged per row, both accountings computed — the deviation moves the pooled rate from 59.0% (first-party only) to 57.5% (mixed) and changes no verdict. Scope after §3.6: the anchoring law is cross-vendor at the pooled level; its full cell-level mode structure remains confirmed only within the original family. Arm V (§3.6) extends the chain under the zero-shot protocol: registered before sampling with three pre-sampling amendments, 877 invocations across thirteen free-tier aliases, ten below the registered floor with their failure taxonomies logged, and the three floor-clearing vendors splitting TRANSFERS/TRANSFERS/DOES-NOT-TRANSFER (`arm_v_score.py`, no arguments, from the live ledger; frozen output `arm_v_score_final.txt`).

Sampling parameters unpinned. The runtime does not expose pinned decoding parameters, so every effect is a distributional claim over the observed sampling regime, and §3.2's mode-ceiling result inherits that scope. This is the subject of the companion paper.

`opus_alias` provenance. The label denotes the serving alias addressed, not an identified model version. Without pinned weights we cannot separate a model-tier effect from a serving-path effect, and the supporting anomalies live only in the runtime transcript.

Confirmatory base for the branch rule, updated by arm M. floor and round disagree exactly on $N \in [k^2 - k + 1, k^2 - 1]$, so every discriminating cell is a trap cell by construction. The truncate arm now has five confirmed k in the square (4, 5, 6, 7, 8 — the last added by arm M's $N = 57$ cell, §3.4) plus two rectangle cells; the extend arm is confirmed only at $m \leq 1$ and **disconfirmed at** $m \geq 4$ (§3.4), which is the sharper scope the abstract now carries. The empirical-k back-out has been run twice as disclosed post-hoc diagnostics: `diagnostics_kmatch.py` (§3.2 — 50/50 on-prediction samples k^*-structured, 64/69 valid overall) and arm M's per-cell k_{emp} distributions (§3.4 — the converge cells sit on k^*+1, which is how the filler-branch failure was diagnosed).

One prompt, one phrasing. Every arm conditions on a single hash-locked prompt and no paraphrase variant was sampled, so the anchor is established as a property of the (model, prompt string) pair; locking buys reproducibility, not robustness to rewording, and prompt-format sensitivity of the size reported by Sclar et al. (2024) is untested here. Relatedly, the cross-vendor concentration differences in §3.6 are measured but not explained; alignment-induced diversity loss (Kirk et al., 2024) predicts vendor-dependent concentration and is a candidate account we did not test.

9 Reproducibility, deviations, and disclosures

Deviations from preregistration are tabled in Appendix B. Provenance, in list form:

- **Registration.** Prompts were hashed with SHA-256 and the hashes recorded before sampling, with the predicted values they were to be tested against: `arm_f_repro.py` (header, P1–P5, square arm), `arm_s_preregistration.txt`, `arm_o_preregistration.txt`, `arm_t_preregistration.txt` (prereg hash `ab7900a8...`), `arm_m_preregistration.txt` with `arm_m_prompts.json` (committed `e528c6b` before sampling), and `arm_mu_preregistration.txt` with `arm_mu_prompts.json` (committed `2b7d202` before sampling; decision thresholds, power analysis, tie conventions and falsifier F-MU1 all registered), and the cross-vendor chain `arm_gm_preregistration.txt` (commit `37b3adb`), `arm_gm2_preregistration.txt` (`3019aab`), `arm_gm3_preregistration.txt` (`c0b5b97`) with its serving-path deviation registered in `arm_gm3_amendment1_openrouter.md` (`67cc4a1`) before the affected cells were sampled, and `arm_v_preregistration.md` (commit `5444f10`, ancestry-checked by its runners before any sampling) with its three amendments, each committed before the sampling it governs. Arm CC's registration (`arm_cc_preregistration.txt` with frozen prompts, executor and smoke test) was committed and pushed to the public remote at `285deca` before its first invocation; arms CC2 and CCS were registered together (`arm_cc2_preregistration.txt`, commit `4d7b7c5`, pushed before sampling) after CC's results were known, with that motivation disclosed in the registration itself. Arm CN (`arm_cn_preregistration.txt`, commit `69a1642`, pushed before sampling) was registered after the rest of the manuscript was complete, in response to a pre-submission check; its motivation is post-hoc with respect to the paper's results and is disclosed in the registration, and its predictions were fixed before its first invocation. The rectangle transfer was registered before collection and never refitted. On multiplicity: this study now spans many registered arms, and no correction across arms is applied — each arm's predictions were registered with its own bars and falsifiers before its sampling, every arm is reported regardless of outcome (three GM-chain instruments failed or stalled before one succeeded, and §3.4's falsifier fired), and no arm was selected for inclusion on its result. The protection against garden-of-forking-paths here is registration and exhaustive reporting, not a family-wise error rate; a reader who wants one can count arms from this section.

- **Four limits on that claim.** (i) The digests cover the task prompt only; the subagent inherits instruction files outside it, so what is locked is a *fragment* of the conditioning context, a replicator cannot reconstruct those files, and we cannot establish they were identical across the arm-F and arm-T windows — the boundary §6.4 identifies as a confound. (ii) `arm_f_prompts.json` covers $N = 13$, 17, 31, 35 and 37; the $N = 21$ bare hash is registered in `arm_t_preregistration.txt`; the $N = 43$ **bare hash appears in no preregistration and no prompts file**, existing only in the ledger — it recomputes exactly from the bare template, but recomputation after the fact is not registration. (iii)

No hash was ever registered for the pilot prompt, whose drift the preregistration describes only in prose; the ten pilot rows carry `prompt_sha256: null` in the corrected ledger with an explanatory note. (iv) The rectangle ledger (`arm_g_candidates.jsonl`) carries no prompt hashes, no run dates and no proposer fields at all.

- **Ledger metadata correction.** Three internal-check findings established that the shipped ledger's provenance fields contradicted the paper: trace rows carried the *bare* prompt hash for their cell, `run_date` was uniformly `2026-07-30` on all 215 rows, and the Sonnet and `opus_alias` rows were stamped with the Haiku alias and dated id. All three are corrected in `arm_f_candidates_v2.jsonl`; the correction is metadata only — every scored quantity, geometry, raw output, arm label and sample id is byte-identical to v1, verified field by field, each corrected field carrying a sibling `*_v1` field. `arm_f_candidates.jsonl` is unmodified (sha256 `02ecabcd...`); v2 is `9f845f78....` Correction rules, sources, wave-boundary arithmetic and caveats — including that `run_date` has day granularity only — are in `ledger_v2_corrections.md`. Analysis scripts read v2.

- **Excluded and unreleased records.** The five concurrency-cap rejections appear in the ledger in no form, no `excluded_reason` field exists, and the exclusion rule was not preregistered; both rates are reported in §3.3. The `opus_alias` latency and token-count anomalies (§5) come from the runtime session transcript; no latency or token field exists in any released artifact.

- **Analysis artifacts.** Raw outputs are stored verbatim (`arm_f_raw.json`, `arm_f_candidates.jsonl`, `arm_f_candidates_v2.jsonl`, `arm_g_candidates.jsonl`, `arm_m_collect.jsonl` scored by `arm_m_analysis.py`, `arm_mu_collect.jsonl` scored by `arm_mu_analysis.py` with frozen output `arm_mu_results.txt`, and the GM3 ledger `arm_gm_gm3_checkpoint.jsonl` — every row's raw API response verbatim, serving path and provider tagged per row — scored by `arm_gm3_analysis.py` into `arm_gm3_report.json` with both serving-path accountings) and scoring is deterministic and local. Value claims are gated on an independent LP oracle rather than on the closed form being tested: `n_sweep_forecast.py` with `verify_against_lp` checks 83 configurations to within 10^{-9}, and the same oracle produced §4.2's negative result. The faithfulness rubric (`p7_faithfulness_rubric.md`) was frozen by git commit `e181d2a` (2026-08-01 03:56:56) before any rescoring, and all 60 blind labels with justifications are in `p7_blind_labels.json`. `arm_t_analysis.py`, `n_sweep_forecast.json`, `rect_forecast.json`, `p11_mode_baseline.json` and `STATE.md` are included; every table regenerates from raw outputs by a command documented in `HOW_TO_RUN.md`.

- **Use of AI systems.** Claude models wrote the collection harnesses, the scoring, analysis and verification scripts, and drafted the manuscript prose; the human author directed the programme, set the research questions, approved each preregistration before sampling, made the final inclusion, stopping and submission decisions, and is solely responsible for the content. The claim a reader can check is the ordering, not the authorship: each preregistration commit is a git ancestor of the sampling it governs. Because the authoring models share a family with arms under study (§5), what a reader is asked to trust is the released artifact — the scripts, raw ledgers, LP oracle and frozen outputs — not the author.

- **Internal checks.** Before submission the manuscript passed an internal adversarial check: language models of other families (`deepseek-v4-pro`, `deepseek-v4-flash`, Gemini), under written protocols, independently recomputed the paper's statistics from the released ledger; thirty-two claims were corrected as a result, itemized in the supplementary `corrections_ledger.md`. This is quality assurance, not peer review, and carries no external authority.

- **Stopping rule.** The manuscript closed after implementing the internal-check findings; no further data-dependent analysis was performed. Arms CC, CC2 and CCS (§3.7) and CN (§3.8) were each registered, run and scored once under their own preregistrations after that close; each analysis script was committed before its sampling and not modified after. Analyses named but not run — the one-parent mutation arm, the recipe-family choice probe, the empirical-k back-out, faithfulness split by on- versus off-prediction, a library-enabled code probe (`numpy/scipy`; the math-only code channel has since run as arm CC), interleaved re-collection of the arm-T pair, and P5 at $n = 20$

— are recorded in §6.5 and §8 as future work, so that the reported statistics remain the ones the preregistrations governed.

References

Sahar Abdelnabi and Ahmed Salem. The hawthorne effect in reasoning models: Evaluating and steering test awareness, 2025. URL https://arxiv.org/abs/2505.14617.

Iván Arcuschin, Jett Janiak, Robert Krzyzanowski, Senthooran Rajamanoharan, Neel Nanda, and Arthur Conmy. Chain-of-thought reasoning in the wild is not always faithful, 2026. URL https://arxiv.org/abs/2503.08679.

Mislav Balunović, Jasper Dekoninck, Nikola Jovanović, Ivo Petrov, and Martin Vechev. Mathconstruct: Challenging llm reasoning with constructive proofs, 2025. URL https://arxiv.org/abs/2502.10197.

Timo Berthold, Dominik Kamp, Gioni Mexi, Sebastian Pokutta, and Imre Polik. Out-of-the-box global optimization for packing problems: New models and improved solutions, 2026. URL https://arxiv.org/abs/2605.04850.

Siddharth Boppana, Annabel Ma, Max Loeffler, Raphael Sarfati, Eric Bigelow, Atticus Geiger, Owen Lewis, and Jack Merullo. Reasoning theater: Disentangling model beliefs from chain-of-thought, 2026. URL https://arxiv.org/abs/2603.05488.

Herbie Bradley, Andrew Dai, Hannah Teufel, Jenny Zhang, Koen Oostermeijer, Marco Bellagente, Jeff Clune, Kenneth Stanley, Grégory Schott, and Joel Lehman. Quality-diversity through ai feedback, 2023. URL https://arxiv.org/abs/2310.13032.

Nicholas Carlini, Florian Tramèr, Eric Wallace, Matthew Jagielski, Ariel Herbert-Voss, Katherine Lee, Adam Roberts, Tom Brown, Dawn Song, Úlfar Erlingsson, Alina Oprea, and Colin Raffel. Extracting training data from large language models. In *30th USENIX Security Symposium*, 2021.

Mert Cemri, Shubham Agrawal, Akshat Gupta, Shu Liu, Audrey Cheng, Qiuyang Mang, Ashwin Naren, Lutfi Eren Erdogan, Koushik Sen, Matei Zaharia, Alex Dimakis, and Ion Stoica. Adaevolve: Adaptive llm driven zeroth-order optimization, 2026. URL https://arxiv.org/abs/2602.20133.

Ziyu Chen, Yilun Zhao, and Arman Cohan. Measuring the gap between human and llm research ideas, 2026. URL https://arxiv.org/abs/2607.01233.

Erich Friedman. Packing center: circles in squares, maximizing the sum of radii. https://erich-friedman.github.io/packing/cirRsqu/, n.d. Best-known values transcribed into n_sweep_forecast.py; accessed 2026.

Yonatan Gideoni, Sebastian Risi, and Yarin Gal. Simple baselines are competitive with code evolution, 2026. URL https://arxiv.org/abs/2602.16805.

Shahriar Golchin and Mihai Surdeanu. Time travel in LLMs: Tracing data contamination in large language models. In *International Conference on Learning Representations (ICLR)*, 2024. arXiv:2308.08493.

Qingyan Guo, Rui Wang, Junliang Guo, Bei Li, Kaitao Song, Xu Tan, Guoqing Liu, Jiang Bian, and Yujiu Yang. Connecting large language models with evolutionary algorithms yields powerful prompt optimizers. In *International Conference on Learning Representations (ICLR)*, 2024. arXiv:2309.08532.

Can Gurkan, Forrest Stonedahl, and Uri Wilensky. Mutation without variation: Convergence dynamics in llm-driven program evolution, 2026. URL https://arxiv.org/abs/2606.05408.

Julien Herrmann and Guillaume Pallez. An in-depth study of llm contributions to the bin packing problem, 2026. URL https://arxiv.org/abs/2510.27353.

Jordan Hoffmann, Sebastian Borgeaud, Arthur Mensch, Elena Buchatskaya, Trevor Cai, Eliza Rutherford, Diego de Las Casas, Lisa Anne Hendricks, Johannes Welbl, Aidan Clark, Tom Hennigan, Eric Noland, Katie Millican, George van den Driessche, Bogdan Damoc, Aurelia Guy, Simon Osindero, Karen Simonyan, Erich Elsen, Jack W. Rae, Oriol Vinyals, and Laurent Sifre. Training compute-optimal large language models, 2022. URL `https://arxiv.org/abs/2203.15556`.

Yiming Huang, Biquan Bie, Zuqiu Na, Weilin Ruan, Songxin Lei, Yutao Yue, and Xinlei He. Understanding the anchoring effect of llm with synthetic data: Existence, mechanism, and potential mitigations, 2026. URL `https://arxiv.org/abs/2505.15392`.

Haozhe Jia. Hindsightbench: A black-box behavioral audit protocol for parametric hindsight in time-indexed llm decision tasks, 2026. URL `https://arxiv.org/abs/2607.18867`.

Liwei Jiang, Yuanjun Chai, Margaret Li, Mickel Liu, Raymond Fok, Nouha Dziri, Yulia Tsvetkov, Maarten Sap, Alon Albalak, and Yejin Choi. Artificial hivemind: The open-ended homogeneity of language models (and beyond), 2025. URL `https://arxiv.org/abs/2510.22954`.

Jared Kaplan, Sam McCandlish, Tom Henighan, Tom B. Brown, Benjamin Chess, Rewon Child, Scott Gray, Alec Radford, Jeffrey Wu, and Dario Amodei. Scaling laws for neural language models, 2020. URL `https://arxiv.org/abs/2001.08361`.

Sourena Khanzadeh. Project ariadne: A structural causal framework for auditing faithfulness in llm agents, 2026. URL `https://arxiv.org/abs/2601.02314`.

Valentin Khrulkov, Andrey Galichin, Denis Bashkirov, Dmitry Vinichenko, Oleg Travkin, Roman Alferov, Andrey Kuznetsov, and Ivan Oseledets. Gigaevo: An open source optimization framework powered by llms and evolution algorithms, 2025. URL `https://arxiv.org/abs/2511.17592`.

Jinwoong Kim, Rui Yang, and Huishuai Zhang. Geobuildbench: A benchmark for interactive and executable geometry construction from natural language, 2026. URL `https://arxiv.org/abs/2605.13167`.

Robert Kirk, Ishita Mediratta, Christoforos Nalmpantis, Jelena Luketina, Eric Hambro, Edward Grefenstette, and Roberta Raileanu. Understanding the effects of RLHF on LLM generalisation and diversity. In *International Conference on Learning Representations (ICLR)*, 2024. arXiv:2310.06452.

Robert Tjarko Lange, Yuki Imajuku, and Edoardo Cetin. Shinkaevolve: Towards open-ended and sample-efficient program evolution, 2025. URL `https://arxiv.org/abs/2509.19349`.

Joel Lehman and Kenneth O. Stanley. Abandoning objectives: Evolution through the search for novelty alone. *Evolutionary Computation*, 19(2):189–223, 2011.

Joel Lehman, Jonathan Gordon, Shawn Jain, Kamal Ndousse, Cathy Yeh, and Kenneth O. Stanley. Evolution through large models, 2022. URL `https://arxiv.org/abs/2206.08896`.

Pan Li. Dictionaries, not darwin: Set-level selection beats llm evolution in scientific equation discovery, 2026. URL `https://arxiv.org/abs/2607.04108`.

Yubo Li, Ramayya Krishnan, and Rema Padman. The chain holds, the answer folds: Trace-answer dissociation in reasoning models under adversarial pressure, 2026. URL `https://arxiv.org/abs/2605.29087`.

Fei Liu, Xialiang Tong, Mingxuan Yuan, Xi Lin, Fu Luo, Zhenkun Wang, Zhichao Lu, and Qingfu Zhang. Evolution of heuristics: Towards efficient automatic algorithm design using large language model. In *International Conference on Machine Learning (ICML)*, 2024. arXiv:2401.02051.

Jiaxu Lou and Yifan Sun. Anchoring bias in large language models: An experimental study, 2024. URL `https://arxiv.org/abs/2412.06593`.

Simon Malberg, Roman Poletukhin, Carolin M. Schuster, and Georg Groh. A comprehensive evaluation of cognitive biases in llms, 2025. URL `https://arxiv.org/abs/2410.15413`.

Elliot Meyerson, Mark J. Nelson, Herbie Bradley, Adam Gaier, Arash Moradi, Amy K. Hoover, and Joel Lehman. Language model crossover: Variation through few-shot prompting, 2024. URL https://arxiv.org/abs/2302.12170.

Jean-Baptiste Mouret and Jeff Clune. Illuminating search spaces by mapping elites. *arXiv preprint arXiv:1504.04909*, 2015.

Alexander Novikov, Ngân Vũ, Marvin Eisenberger, et al. Alphaevolve: A coding agent for scientific and algorithmic discovery, 2025. URL https://arxiv.org/abs/2506.13131.

OpenAI. OpenAI o3 and o4-mini system card. https://openai.com/index/o3-o4-mini-system-card/, 2025.

Bernardino Romera-Paredes, Mohammadamin Barekatain, Alexander Novikov, Matej Balog, M. Pawan Kumar, Emilien Dupont, Francisco J. R. Ruiz, Jordan S. Ellenberg, Pengming Wang, Omar Fawzi, Pushmeet Kohli, and Alhussein Fawzi. Mathematical discoveries from program search with large language models. *Nature*, 625(7995):468–475, 2024. doi: 10.1038/s41586-023-06924-6.

Daniel Scalena, Sara Candussio, Luca Bortolussi, Elisabetta Fersini, Malvina Nissim, and Gabriele Sarti. Beyond the commitment boundary: Probing epiphenomenal chain-of-thought in large reasoning models, 2026. URL https://arxiv.org/abs/2606.13603.

Melanie Sclar, Yejin Choi, Yulia Tsvetkov, and Alane Suhr. Quantifying language models' sensitivity to spurious features in prompt design or: How i learned to start worrying about prompt formatting. In *International Conference on Learning Representations (ICLR)*, 2024. arXiv:2310.11324.

Asankhaya Sharma. Openevolve: an open-source implementation of AlphaEvolve. GitHub repository, https://github.com/codelion/openevolve, 2025.

Parshin Shojaee, Iman Mirzadeh, Keivan Alizadeh, Maxwell Horton, Samy Bengio, and Mehrdad Farajtabar. The illusion of thinking: Understanding the strengths and limitations of reasoning models via the lens of problem complexity, 2025. URL https://arxiv.org/abs/2506.06941.

Kevin Sim, Quentin Renau, and Emma Hart. Beyond the hype: Benchmarking llm-evolved heuristics for bin packing, 2025. URL https://arxiv.org/abs/2501.11411.

Charlie Snell, Eric Wallace, Dan Klein, and Sergey Levine. Predicting emergent capabilities by finetuning, 2024. URL https://arxiv.org/abs/2411.16035.

Chang Su, Zhongkai Hao, Zhizhou Zhang, Zeyu Xia, Youjia Wu, Hang Su, and Jun Zhu. Helix: Evolutionary reinforcement learning for open-ended scientific problem solving, 2026. URL https://arxiv.org/abs/2603.07642.

Zhi Rui Tam, Cheng-Kuang Wu, Yi-Lin Tsai, Chieh-Yen Lin, Hung yi Lee, and Yun-Nung Chen. Let me speak freely? a study on the impact of format restrictions on performance of large language models, 2024. URL https://arxiv.org/abs/2408.02442.

Maria Thomas, Kristina Gligoric, and Nihar B. Shah. Mitigating llm-based p-hacking by preregistering for the next llm, 2026. URL https://arxiv.org/abs/2606.27687.

Michelle Vaccaro. Preregistration for experiments with ai agents, 2026. URL https://arxiv.org/abs/2606.11217.

Niki van Stein and Thomas Bäck. Llamea: A large language model evolutionary algorithm for automatically generating metaheuristics, 2025. URL https://arxiv.org/abs/2405.20132.

Haorui Wang, Parshin Shojaee, Kazem Meidani, Kunyang Sun, José Miguel Hernández-Lobato, Teresa Head-Gordon, Jiajun He, Chandan K. Reddy, Chao Zhang, and Yuanqi Du. Towards diverse scientific hypothesis search with large language models, 2026. URL https://arxiv.org/abs/2606.10587.

Yiping Wang, Shao-Rong Su, Zhiyuan Zeng, Eva Xu, Liliang Ren, Xinyu Yang, Zeyi Huang, Xuehai He, Luyao Ma, Baolin Peng, Hao Cheng, Pengcheng He, Weizhu Chen, Shuohang Wang, Simon Shaolei Du, and Yelong Shen. Thetaevolve: Test-time learning on open problems, 2025. URL `https://arxiv.org/abs/2511.23473`.

Marcus Williams, Hannah Sheahan, Cameron Raymond, Tomek Korbak, Deng Pan, Peilin Yang, Leon Maksin, Ningyi Xie, Phillip Guo, Ian Kivlichan, and Micah Carroll. Predicting llm safety before release by simulating deployment, 2026. URL `https://arxiv.org/abs/2607.07184`.

Yang Wu, Yifan Zhang, Yiwei Wang, Yujun Cai, Yurong Wu, Yuran Wang, Ning Xu, and Jian Cheng. Answer-centric or reasoning-driven? uncovering the latent memory anchor in llms, 2025. URL `https://arxiv.org/abs/2506.17630`.

Xia Yang, Xuanyi Zhang, Hao Hu, and Feng Ji. Beyond accuracy: Evaluating strategy diversity in llm mathematical reasoning, 2026. URL `https://arxiv.org/abs/2605.09292`.

Longfei Yun, Chenyang An, Zilong Wang, Letian Peng, and Jingbo Shang. The price of format: Diversity collapse in llms, 2025. URL `https://arxiv.org/abs/2505.18949`.

Dong Zhang. Testing frontier large language models' physics literacy in parallel physical worlds, 2026. URL `https://arxiv.org/abs/2607.00276`.

Jiayi Zhang, Simon Yu, Derek Chong, Anthony Sicilia, Michael R. Tomz, Christopher D. Manning, and Weiyan Shi. Verbalized sampling: How to mitigate mode collapse and unlock llm diversity, 2026a. URL `https://arxiv.org/abs/2510.01171`.

Rui Zhang and Zhichao Lu. Rethinking code similarity for automated algorithm design with llms, 2026. URL `https://arxiv.org/abs/2603.02787`.

Rui Zhang, Fei Liu, Xi Lin, Zhenkun Wang, Zhichao Lu, and Qingfu Zhang. Understanding the importance of evolutionary search in automated heuristic design with large language models, 2024. URL `https://arxiv.org/abs/2407.10873`.

Xinhao Zhang, Xi Chen, François Portet, and Maxime Peyrard. What makes an llm a good optimizer? a trajectory analysis of llm-guided evolutionary search, 2026b. URL `https://arxiv.org/abs/2604.19440`.

Lexin Zhou, Wout Schellaert, Fernando Martínez-Plumed, Yael Moros-Daval, Cèsar Ferri, and José Hernández-Orallo. Larger and more instructable language models become less reliable. *Nature*, 634(8032):61–68, 2024. doi: 10.1038/s41586-024-07930-y.

A Prompts, verbatim

A.1 Bare template (`{n}` parameterized; the $N = 13$ instance hashes to `32db485bea625ff9f39f4723ebf1a01f337559a9e2cf567fb486928f71f7f8df`):

```
Pack exactly {n} non-overlapping circles inside the unit square [0,1]x[0,1]
so that the sum of their radii is as large as possible.
MUST hold: exactly {n} circles; every circle fully inside the unit square
[0,1]x[0,1] (x-r >= 0, x+r <= 1, same for y); no two circles overlap
(distance between centers >= sum of the two radii; touching is allowed).
Do not write or execute code - construct the packing by reasoning alone.
Output ONLY the raw Python list of {n} [x, y, r] lists. No explanation, no
code fences, no other text.
```

A.2 `trace_v2` template — identical to A.1 through line 3, then:

```
First write one line beginning "METHOD:" naming the construction you used.
Do not use square brackets anywhere in that line.
After the METHOD line, output ONLY the raw Python list of {n} [x, y, r]
lists. No explanation, no code fences, no other text after the list.
```

A.3 Dispatch wrapper (both arms; *not* part of the hashed task prompt, hence an instance of the fragment-hashing limitation in §3.1): `Do not use any tools. Your entire final message must be the answer and nothing else.`

A.4 Registered prompt hashes. bare N=13 `32db485b...`, N=17 `8437df75...`, N=21 `a415425b...`, N=31 `a664d003...`, N=35 `3b08c56e...`, N=37 `2c0a8819...`; trace_v2 N=13 `a920f1c9...`, N=21 `91205727...`, N=31 `bd490b7b...`. Arm CN (held-out N): N=50 `0b858558...`, N=58 `200c0bb4...`, N=62 `b9854a84...`, N=65 `5fb5fefd...`, N=75 `a5db8fc3...`, registered at commit `69a1642`. The N=43 bare hash `1208e7d2...` is ledger-only, and the pilot prompt was never hashed (§9).

A.5 Parser. Completions are parsed with `ast.literal_eval` after stripping code fences; a completion failing to yield a list of N numeric triples is recorded `literal_eval_failed` and scored invalid, with the failure mode logged (§6.2 reports parse and geometric failures separately). Radii must be strictly positive; containment and pairwise non-overlap are checked at the two registered tolerances.

B Deviations and disclosure table

Table 4: Deviations from preregistration.

Prediction	**Registered rule**	**As registered?**	**Outcome / reason**
P1–P3	exact-value point predictions	yes	§3.3, Table 1
P4 (N=37)	converge, 3.0345178	yes	confirmed
P5 (N=35)	truncate, 2.9166667, structural separation	yes	3/4 valid; $n = 4$, "consistent with"
P-S1...P-S4	directional tier comparisons	yes	§5
P-O1, P-O2, P-O4	tier comparisons on on-prediction / rival rates	**no**	declared not evaluable post hoc after a 13% validity collapse; the registered disconfirmation (regression toward the trap) did not occur
P-O3	multi-radius fraction $\geq$ 0.9 of valid	yes	trivially satisfied on 4/4 valid
P-T1	direction at each N **and** pooled $p < 0.05$	yes	not confirmed, $p = 0.30$
P-T2	directional, no alpha	yes	confirmed as registered (1/53 vs 2/50); no inferential weight claimed
P-T3	directional, no alpha	yes	met as registered; $p = 0.0325$ uncorrected fails Holm; carried by $N = 13$ (§6.4)
P-T4	$\geq$ 90% of scoreable claims match	yes, plus corrected scorer	38/41 (93%) original; 54/56 (96.4%) blind, frozen rubric
P-M1–P-M3	modal output = $V(k^*, m)$ at $N =$ 20/30/41	yes	**disconfirmed at all three cells** — modal outputs are (k^*+1)-grid values (§3.4)
P-M4	$N = 57$ modal = $T(8, 57)$, rival count 0	yes	confirmed (6/10 on-prediction, 0/10 rival)
F-M1 (arm M falsifier, filler branch)	modal $\neq V(k^*, m)$ at 2+ of 3 converge cells, ties count as NOT	yes, tie-stated in registration	**triggered at 3 of 3**; $V(k, m)$ restricted to $m \leq 1$ (§3.4)
F-M2 (arm M falsifier, truncate branch)	$N = 57$ modal $\neq T(8, 57)$	yes	not triggered
Arm M rejection rule	runtime deaths excluded, counted	yes (registered for this arm)	3 $N = 41$ invocations excluded, cell reported as 12 of 15

continued on next page

Prediction	**Registered rule**	**As registered?**	**Outcome / reason**
GM3 serving path	first-party API only	amendment registered **before resumption** (`arm_gm3_amendment1_openrouter.md`, `67cc4a1`)	final 41 of 140 cells via routed path, provider logged per row; dual accounting 59.0% / 57.5%, no verdict change (§3.6, §8)
Arm V proposer set	3 opencode aliases	expanded 3 → 13 by amendments 1–2, each registered **before the sampling it governs**	verdicts in §3.6; `big-pickle` disclosed as vendor-unattested in amendment 1
Arm V transport repair	single pass, `max_tokens` 8192	amendment 3 registered **before resumption** (mechanical only: 24,576 budget, retry/backoff, censored-row accounting)	877 attempt rows over 325 slots; floor evaluated per slot (§3.6)
GM3 rival-structure check	—	**post hoc**, disclosed (`diagnostics_gm3_rival.py`)	27/27 rival-valued discriminating-cell packings carry the rival's two-radius structure (§3.6); the registered signature is scoped to k^* and cannot see (k^*-1) structure
Falsifier	trace_v2 validity `<=` bare at 2+ of 3 N	**both readings reported**	**triggered** under the registered tie-inclusive reading (2/3 cells); not triggered under the strict `<` the code implemented (§6.3)
Ladder inclusion	—	**post hoc**	`opus_alias` initially excluded, later included after results known; ladder reported both ways (§5)
Cap exclusion	—	**not registered**	five records excluded; both rates reported (§3.3)